

A review of algorithms for detecting volcanic hot spots in satellite infrared data

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Abstract Since the 1980s, application of thermal infrared satellite data for volcano monitoring has rapidly evolved to become a proven operational tool. Due to the large quantities of data provided by sensors in polar and geostationary orbits, as well as the sheer number of active volcanoes on earth, processing and managing such data sets requires an enormous amount of workforce. A number of algorithms have been developed to facilitate detection, location, and tracking of hot spots of active volcanoes. A collation and review of hot spot detection algorithms developed and applied by the volcanological community reveals three main types which have been applied to date: contextual, fixed threshold, and temporal. The founding algorithms for these three classes are VAST, MODVOLC, and RST, respectively. Through comparison with manually based detections, the performance of each algorithm was tested for sustained lava flows (Etna and Stromboli), strombolian activity (Stromboli), lava dome growth and collapse (Augustine), and fumarole fields (Vulcano). It is shown that, as the number of correctly identified anomalies increases, so too does the number of false positives. Although each of the algorithms operates well within the limits and criteria of their design requirements and

application, under current data restraints, no algorithm can be expected to perform perfectly.

Keywords Hot spot detection algorithms · Thermal remote sensing · Volcano monitoring

Introduction

A series of papers published during the 1980s showed how thermal infrared data acquired by satellite-based sensors could be used to detect, quantify, and track thermal radiance from active volcanic phenomena (Francis and Rothery 1987; Glaze et al. 1989; Rothery et al. 1988; Weisnet and D' Aguanno 1982). Building on this initial work, studies during the early- to mid-1990s began to address the issue of volcano hot spot detection in satellite data (Harris et al. 1995; Oppenheimer and Rothery 1991; Wooster and Rothery 1997a) as well as automated hot spot detection (Dean et al. 1998; Harris et al. 1995; Tramutoli 1998; Wooster and Rothery 1997b). Initial automated detection methods focused on low spatial resolution (1.1 km pixel), but high temporal resolution (one image every 6 h), data provided by the advanced very high resolution radiometer (AVHRR) (Harris et al. 1995; Tramutoli 1998) and the along track scanning radiometer (Wooster and Rothery 1997a). At the same time, hot spot detection algorithms began to be applied to data acquired at lower spatial resolution (4 km²), but higher temporal resolution (15–30 min), as provided by NOAA's Geostationary Operational Environmental Satellite (GOES) sensor (Harris et al. 1997b). With the launch of the moderate resolution imaging spectroradiometer (MODIS) in 1999, algorithms extended to include the 1 km global data base provided by this sensor (Flynn et al. 2002; Wright et al. 2002).

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The global, synoptic, and repeat capability offered by orbiting infrared sensors are useful in monitoring volcanoes on a local (e.g., Calvari et al. 2005) through regional (e.g., Rothery et al. 2005) to global scale (e.g., Wright et al. 2004). Such a capability is especially advantageous in areas where other geophysical or observational monitoring methods are not possible due to logistical reasons, remoteness or sheer number, and extent of active volcanoes (e.g., Dean et al. 2004; Dehn et al. 2000; Patrick et al. 2005). An underlying motivation for one direction of volcano thermal remote sensing research has thus focused on the generation of algorithms capable of routine and timely detection of hot spots for volcanic monitoring purposes, as well as to allow efficient inquiry of temporally and spatially extensive data sets. Our aim is to review the three main types of algorithms that have been designed and developed to tackle this problem and to assess the applicability and accuracy of each algorithm type for detecting volcanic activity across a variety of hot spots scenarios.

Background: suitable data, detection principles, and technique classification

Two portions of the electromagnetic spectrum have been used for volcano hot spot detection: (1) the mid-infrared (MIR), specifically the atmospheric window between 3.5 and 4 μm , and (2) the thermal infrared (TIR), with the 10 to 12 μm window typically being used. Two commonly used sensors which provide data spanning these wavebands are AVHRR and MODIS. These sensors also provide acceptable temporal resolutions for monitoring purposes, allow global coverage (including polar regions) and provide digital data directly to the user. This latter capability is essential if data are to be processed in real time to allow rapid response.

AVHRR is flown on the polar orbiting NOAA satellite series which, as of January 2009, had six satellites in orbit (NOAA-10 plus NOAA-14 through -19). The AVHRR sensors have five spectral bands with a nominal spatial resolution of 1.1 km at nadir (Cracknell 1997). The bands most commonly utilized in detecting volcano, fire, and industrial-related hot spots are located in the MIR (band 3) and the TIR (bands 4 and 5) (Table 1). Policy is to maintain at least two satellites in orbit at any one time; this allows for multiple views of any volcano during any given day. Currently, six operational satellites multiplied by two passes

per day results in 12 images acquired per day for any equatorial point, increasing toward the poles due to orbit convergence (see Mouginiis-Mark and Domergue-Schmidt 2000 for full review of orbital characteristics for satellites typically used for volcanological applications).

MODIS is flown on NASA's EOS polar orbiting satellites (Terra and Aqua). MODIS has 36 spectral bands, 16 of which are located in the MIR and TIR (Table 2). MIR bands 22 and 21, plus TIR band 32, are typically used for hot spot detection. The advantage of band 21 is that it has a high gain setting, having a saturation temperature (maximum detection limit) of $\sim 240^\circ\text{C}$. Like the NOAA satellites, both Terra and Aqua have two equatorial crossings per day, allowing two images to be acquired during a 24-h period: one acquired during the daytime portion of the orbit and one acquired during the nighttime portion. With two satellites in orbit, this allows at least four images to be acquired over any sub-aerial volcano each day.

Detection principles

Most hot spot detection techniques are based on two related principles: Wien's displacement law and the Planck function. Peak spectral radiance emitted by a high-temperature body (e.g., those at magmatic and lava temperatures) occurs in the short-wave infrared and MIR. In contrast, for bodies at earth ambient temperatures the point of peak emission occurs in the TIR (Fig. 1a). This shift of peak emission to longer wavelengths with decreased temperature is described by Wien's displacement law:

$$\lambda_{\text{max}} = A/T \quad (1)$$

A being Wien's displacement constant (2,898 $\mu\text{m K}$). For blackbodies radiating at 1,000°C, 500°C, and 20°C, this defines peak emission wavelengths of 2.28, 3.75, and 9.89 μm , respectively (Fig. 1a). The Planck function describes how radiation emitted by the surface varies with temperature (T) and wavelength (λ), and is given by:

$$M(\lambda, T) = 2\pi hc^2 \lambda^{-5} \left[\exp\left(\frac{hc}{\lambda kT}\right) - 1 \right]^{-1} \quad (W m^{-2} m^{-1}) \quad (2)$$

Here, h is Planck's constant (6.6256×10^{-34} J s), c is the speed of light (2.9979×10^8 m/s), and k is the Boltzmann gas constant (1.38×10^{-23} J K $^{-1}$). This function, as plotted in Fig. 1a, shows that hotter bodies emit more radiation

Table 1 AVHRR sensor characteristics

EM region	Band	Wavelength (μm)	Spatial resolution (km^2)	Saturation temperature ($^\circ\text{C}$)
MIR	3	3.55–3.93	1.1	~ 50
TIR	4	10.3–11.3	1.1	~ 70
	5	11.4–12.4	1.1	~ 70

Table 2 MODIS sensor characteristics

EM region	Band	Wavelength (μm)	Spatial resolution (km^2)	Saturation temperature ($^{\circ}\text{C}$)
MIR	21	3.929–3.989	1.0	~227
	22	3.929–3.989	1.0	~60
TIR	31	10.78–11.28	1.0	~115
	32	11.77–12.27	1.0	~115

than cooler bodies at all wavelengths. However, the difference between the radiance emitted by the two bodies differs with wavelength, being much greater in the MIR than in the TIR.

As initially shown by Doizer (1981), the strong emission from a high-temperature body in the MIR means that the presence of a sub-pixel hot spot will cause the pixel-integrated radiances in the MIR to be much greater than for the same pixel in the TIR. In such a case, the integrated radiance at wavelength λ (R_λ) can be described by:

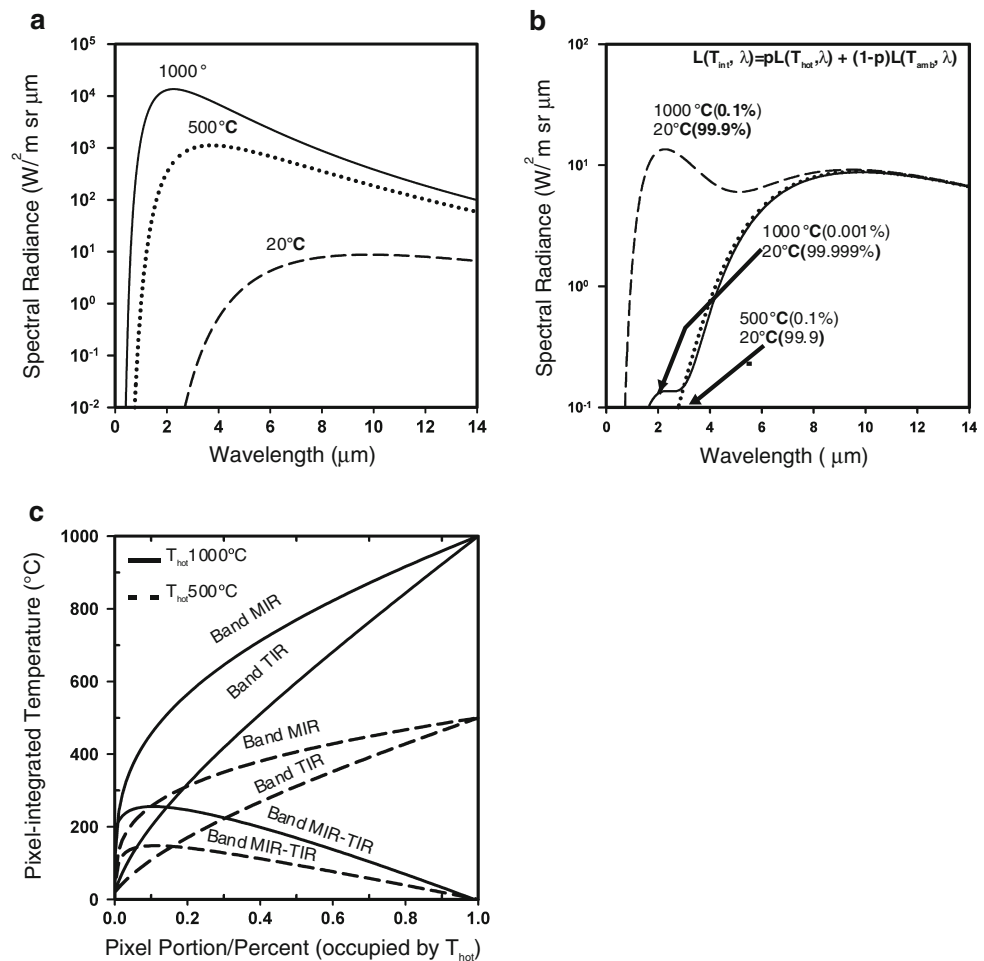
$$M(\lambda, T) = R_\lambda = p R_h + (1 - p) R_b \quad (3)$$

R_h and R_b being the spectral radiance of the sub-pixel hot spot and the surrounding background, and p being the pixel

portion occupied by the hot spot. This effect means that, as the size and temperature of the sub-pixel hot spot increases, a much larger increase in the integrated radiance is experienced in the MIR than in the TIR (Fig. 1b). For example, the difference in radiance between a pixel with a $1,000^{\circ}\text{C}$ hot spot occupying 0.001% of the pixel and one with 0.1% at $1,000^{\circ}\text{C}$ is minimal at $11 \mu\text{m}$, being $8.48 \text{ Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$ and $8.73 \text{ Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$ for the two cases, respectively. However, there is a large difference at $3.5 \mu\text{m}$, the two cases now yielding $0.27 \text{ Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$ and $9.13 \text{ Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$, respectively (Fig. 1b).

Because spectral radiances are used to calculate pixel-integrated temperatures through inversion of Eq. 2, this effect means that pixel-integrated temperatures for pixels

Fig. 1 Planck and pixel-integrated temperature curves for a variety of volcanic scenarios. **a** Planck curves for blackbodies at $1,000^{\circ}\text{C}$, 500°C , and 20°C . **b** Planck curves for mixed pixels composed of two components, one at ambient background (20°C) and the second by a surface at $1,000^{\circ}\text{C}$ and occupying 0.001% to 0.1% of the pixel or at 500°C and occupying 0.1%. **c** Change in pixel-integrated temperature with percent coverage by a hot source for mixed pixels composed of two components. The first source is at ambient (20°C) and the second at $1,000^{\circ}\text{C}$ or 500°C . Temperatures are shown for the MIR band, the TIR band and the difference between the two (MIR–TIR)



containing sub-pixel hot spots will be higher in the MIR than in the TIR. The difference between pixel-integrated temperature in the MIR (T_{MIR}) and TIR (T_{TIR}) is described by $\Delta T (= T_{\text{MIR}} - T_{\text{TIR}})$ and thus also depends on the size and temperature of the sub-pixel heat source (Fig. 1c). For example, the pixel with a 1,000°C source occupying 0.1% of the pixel will have a pixel-integrated temperature of 118°C in the MIR and 58°C in the TIR, so that $\Delta T=60^\circ\text{C}$. In contrast, for pixels which do not contain sub-pixel hot spots T_{MIR} will equal T_{TIR} (Fig. 1c). This principle was the foundation of initial hot spot detection algorithms, being introduced by the fire community in multi-step, fixed threshold detection algorithms proposed by, for example, Flannigan and Vonder Haar (1986), Kennedy et al. (1994) and Arino and Melinotte (1995). It is particularly effective in isolating hot spots from solar heated backgrounds on single images, where heating of vegetation-free lava and scoria fields can raise surface temperatures to 60–70°C, which in the above example will cause the pixel-integrated temperature of the hot spot containing pixel to be equal to or less than that of the background. However, theoretically and given the same emissivity in the MIR and TIR, consideration of ΔT should reduce all background pixels to a value of around 0°C and highlight the hot spot containing pixel (Harris et al. 1997a).

Classification of volcano hot spot detection techniques

Depending on whether a hot spot detection technique operates by assessing a pixel's radiance (or temperature) in a spectral, spatial or temporal space, hot spot detection methods can be divided into three categories: (1) fixed threshold, (2) contextual, or (3) temporal. Fixed threshold algorithms are spectrally based and use data for a single pixel to assess whether the radiance (or temperature) in the MIR and/or TIR bands is thermally anomalous or not. The primary algorithm in this class is the MODVOLC algorithm of Flynn et al. (2002) and Wright et al. (2002a). Contextual algorithms use the difference between a pixel's radiance (or temperature) and that of its surrounding pixels to assess whether it contains a hot spot or not, the original algorithm in this type being the VAST method of Harris et al. (1995). Temporal algorithms operate by comparing a pixel's radiance (or temperature) with mean values obtained for the same pixel from time series of data. These time series allow typical pixel values for any given time of day and year to be defined, and divergences from the norm to be statistically assessed, with the core method of this type in volcanology being the RAT approach of Tramutoli (1998). As shown in Fig. 2a, these algorithms form the points of a ternary classification. However, the Fig. 2a classification shows that, while most algorithms can be tied to one of the three main algorithm types, three (MODLEN, Okmok, and

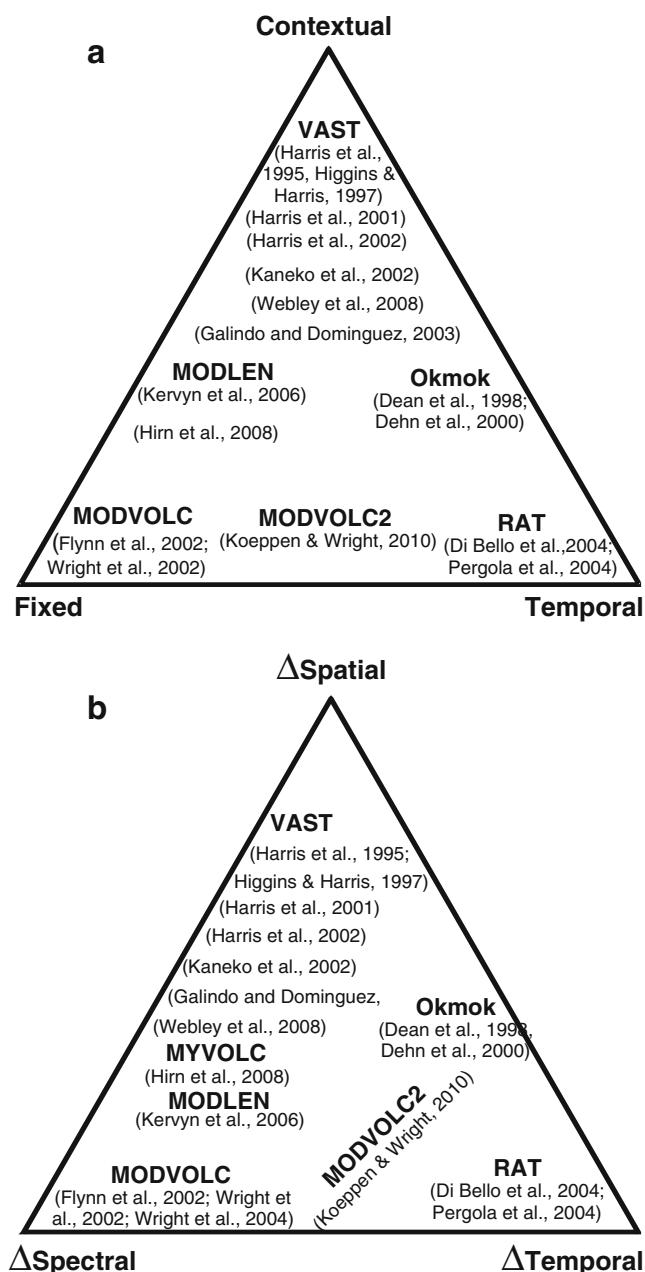


Fig. 2 Classification of volcanic hot spot detection algorithms. **a** Classification based on algorithm type (see text for definition) and **b** classification based the spatial, spectral, and temporal principles each algorithm employs to detect thermal anomalies

MODVOLC2) incorporate aspects of one or more of the three types and therefore reside between the end-members.

Automated algorithms

History of algorithms

Fixed threshold algorithms were first developed in the 1980s by the wildfire community and used AVHRR data to automatically detect fires (e.g., Flannigan and Vonder Haar

1986; Kaufman et al. 1990; Kennedy et al. 1994). They operated by setting a threshold value and flagged any value which exceeded that threshold. For example, Kaufman et al. (1990) ran three tests to find thermally anomalous pixels: (1) $T_{\text{MIR}} \geq 316$ K, (2) $\Delta T \geq 10$ K, and (3) $T_{\text{TIR}} > 250$ K. These tests, respectively, ensured that (1) the pixel was hot, (2) a sub-pixel hot spot was present, and (3) the pixel was cloud free. Each algorithm varied the level of the thresholds set, setting different thresholds depending on the region monitored. Such algorithms performed well but were limited to the specific regions for which the thresholds were set (Justice and Dowty 1994).

A volcano-targeted single-step fixed threshold algorithm was tested by Di Bello et al. (2004) on images of NE Sicily containing Etna. They showed that such a simple single-step threshold of $\Delta T > 10$ K produced hundreds of false detections, making the misleading claim that this failure-of-an-algorithm was the contextual approach of Harris et al. (1995). It is not. The first, robust, fixed threshold approach applied to volcano hot spot detection was that of Harris et al. (2002a) who introduced a multi-step threshold algorithm to detect hot spots in whole earth disc images acquired by GOES. This method was introduced because application of the initial contextual algorithm used to detect hot spots in GOES data for just the Hawai'i sector (Harris et al. 2001) was unworkable when applied to the whole GOES image. Instead, a series of tests that used the reflective, ΔT , MIR, and TIR properties of the pixel and its background were used to assign a probability of the pixel containing a hot spot (see Table 3 of Harris et al. 2002b). If the probability exceeded a threshold value, then the pixel was flagged as potentially hot and assigned a gray scale depending on the probability.

MODVOLC (Flynn et al. 2002; Wright et al. 2002; Wright et al. 2004) is another fixed threshold algorithm. Developed at the Hawai'i Institute of Geophysics and Planetology (HIGP), it is designed to detect volcanic thermal anomalies on a global scale using MODIS data under limited operational constraints (Wright et al. 2002). The algorithm went operational in February 2000, running at the NASA DAAC shortly after the December 1999 launch of MODIS on NASA's Terra satellite.

The output has subsequently been widely used, contributing to activity reports by the Smithsonian Institute's Global Volcanism Network, as well as to identify hot spots and produce radiance time series during effusive episodes at, for example, Anatahan (Wright et al. 2005), Mt. Belinda (Patrick et al. 2005), Melanesia (Rothery et al. 2005), Stromboli (Ripepe et al. 2005), Semeru and Merapi (Harris and Ripepe 2007a), and Fuego (Lyons et al. 2010).

Contextual algorithms were also first developed by the wildfire community during the 1990s (e.g., Flasse and Ceccato 1996; Justice and Dowty 1994; Langaas 1993; Lee and Tag 1990). These approaches applied flexible thresholds based upon statistics calculated from pixels in the image region immediately surrounding a target pixel. For example, the fire detection algorithm of Lee and Tag (1990) executed five steps that used AVHRR MIR and TIR data to determine whether or not the brightness temperature of a target pixel was anomalously elevated above that of its eight neighboring pixels. Such a contextual approach was the basis of the first automated volcanic hot spot detection algorithm applied in volcanology as reported by Harris et al. (1995). Inspired by the work of Lee and Tag (1990) and Langaas (1993), the algorithm was later developed into the Volcanic Anomaly Software (VAST) algorithm of Higgins and Harris (1997). Initially, VAST was tested using AVHRR data for Etna, Vulcano, Stromboli, and Lipari (Harris et al. 1995; Higgins and Harris 1997), as well as wildfires in Australia (Harris 1996), and was designed to detect thermally anomalous pixels by comparing the ΔT for each pixel with the mean from its surrounding 8 pixels. The algorithm was adapted to run on GOES data for Hawai'i by Harris et al. (2001c), whereby any pixel with ΔT that was greater than the ΔT mean plus 3.3σ for a 5-pixel-wide box surrounding a 10×10 pixel target zone was flagged as a hot spot. Since VAST's development, other algorithms have been designed based on the similar principles and operations (Fig. 2a, b). Many of these algorithms have been implemented for local-to-regional monitoring purposes including AVHRR-based monitoring the volcanoes of Mexico (Galindo and Dominguez 2002), Central America (Webley et al. 2008), and Japan (Kaneko et al. 2002).

Table 3 Test cases of volcanic activity and satellite imagery used in this study

Case	Location	Eruption date	Data range	AVHRR		MODIS	
				Images	Cloud free	Images	Cloud free
Sustained lava flows	Etna	9/11/04–03/25/05	9/1/04–03/11/05	349	249	697	262
Short flows	Stromboli	2/27/07–4/2/07	2/25/07–4/11/07	181	22	118	19
Strombolian activity	Stromboli	N/A	1/1/06–12/31/06	899	386	1,707	468
Lava dome growth and collapse	Augustine	01/11/06–3/27/06	01/01/06–3/31/06	1,230	297	528	139
Fumarolic activity	Vulcano	N/A	01/1/06–12/1/06	893	324	1,707	456
Null	Lipari	N/A	01/1/06–12/1/06	893	319	1,707	460

The last algorithm type was developed to detect hot spots using temporal variations in pixel brightness temperature. Tramutoli (1998) originally proposed the technique, named Robust AVHRR Technique (RAT) to minimize the effects that different geographical and seasonal conditions can have on some fixed threshold and contextual algorithms, proposing the use of RAT in a variety of applications including detection of volcanic ash clouds, dust storms, forest fires, and earthquake-related hot spots. To achieve this, the algorithm looks through a data archive to see how the temperature for a specific pixel changes through time, thereby assessing variation caused by seasonal and atmospheric effects. By comparing the pixel's current temperature to the past temperature history of that pixel, divergence from the "norm" can be detected. Given that exactly the same pixel needs to be considered through time, the technique requires precise, sub-pixel location accuracy, and excellent image-to-image registration (Tramutoli et al. 2001; Pergola and Tramutoli 2003). Pergola et al. (2001), using AVHRR images for Etna, subsequently showed how the method could be used with reflected (AVHRR band 1) data to detect ash clouds, developing an index named the Statistical Normalized Albedo Excess. At the same time, Pergola et al. (2001) developed the Absolutely Local Index of Change of the Environment (ALICE) that used TIR (AVHRR band 4) data to detect thermal anomalies associated with seismic activity. Finally, Pergola et al. (2004) and Di Bello et al. (2004) showed how RAT could be used with MIR (AVHRR band 4) data to detect hot spots associated with lava flow activity on Mt. Etna through application of the ALICE index. Since that time the algorithm has been used with other satellite data and has been renamed Robust Satellite Technique (RST) (Pergola et al. 2009; Pergola et al. 2008). More recently, Yuhaniz and Vladimirova (2009) have proposed a similar change detection technique that uses tiling of cross-referenced image time series and a "fuzzy inference engine" to assess the degree of change in a given pixel and issue alerts if a major change occurs.

MODVOLC

MODVOLC is based on the ΔT principle and implements a Normalized Thermal Index (NTI) to distinguish pixels containing sub-pixel hot spots (Flynn et al. 2002; Wright et al. 2002). The NTI considers the difference in radiance between MODIS bands 22 (MIR) and 32 (TIR) and then divides (normalizes) this by the sum of the radiances in the same two bands:

$$NTI = \frac{L_{22} - L_{32}}{L_{22} + L_{32}} \quad (4)$$

This gives an NTI in the range of -1.0 to 1.0 . Several tests were run to determine an NTI threshold that returned the minimum incidence of false positives when run on a

global data set. This was found to be -0.8 for nighttime data, so that any pixel with an $NTI \geq 0.8$ was flagged as anomalous in the global hot spot searches of Flynn et al. (2002) and Wright et al. (2002a). The algorithm executes seven steps as given in Fig. 3. Step 1 is image acquisition, followed by a data-quality check. Step 2 determines if band 22 is saturated, and step 3 computes the NTI (using band 22, or band 21 if band 22 is saturated). Step 4 checks whether the NTI is greater than -0.8 . If so, step 5 writes the hot spot details to the MODVOLC data base. Step 6 then sends the pixel information (location and radiance) to HIGP via ftp, with the information being archived and posted on the MODVOLC website (<http://modis.higp.hawaii.edu>) as part of step 7.

Because the MODVOLC threshold is set at a high level, necessary to allow global detection with minimal false positives, it is not designed to detect cooler, more subtle thermal anomalies. Thus, Kervyn et al. (2008) modified MODVOLC to permit detection of lower amplitude thermal anomalies associated with the effusive carbonatite activity of Oldoinyo Lengai. This locally modified algorithm (MODLEN) involved defining a lower NTI threshold of -0.83 , and adding a contextual step whereby, if the pixels did not meet this threshold, the NTI was also compared with that of its eight neighboring pixels to assess whether an anomaly existed or not (Kervyn et al. 2008). Oldoinyo Lengai, with Coppola et al. (2009) also carrying out a regional adjustment to the NTI to allow detection of hot pixels and construction of radiance time series for Piton de la Fournaise.

Volcanic Anomaly Software

VAST executes eight steps (Fig. 3). Step 1 involves extracting a 30×30 pixel sub-image centered on the volcano of interest. Step 2 then defines two regions: a central volcanic region (containing the target volcano) and non-volcanic region surrounding the volcanic region. In step 3, a ΔT image is created, with step 4 involving calculation of ΔT_{diff} for each pixel, this being the difference in ΔT between a pixel and the mean ΔT of its eight neighboring pixels. In step 5, the non-volcanic portion of the image is scanned for the highest ΔT_{diff} . In step 6, the threshold is applied, this being: if ΔT_{diff} for any pixel in the volcanic region exceeds the maximum encountered in the non-volcanic region, then the pixel is flagged as anomalous (Harris et al. 1995; Higgins and Harris 1997). VAST was subsequently used by Vicari et al. (2009) to locate hot spots in MODIS data during Etna's 2006 eruption and convert to time-average discharge rates to contribute to lava flow simulations.

An updated version of VAST is now available. Although the updated version uses the same contextual approach, it

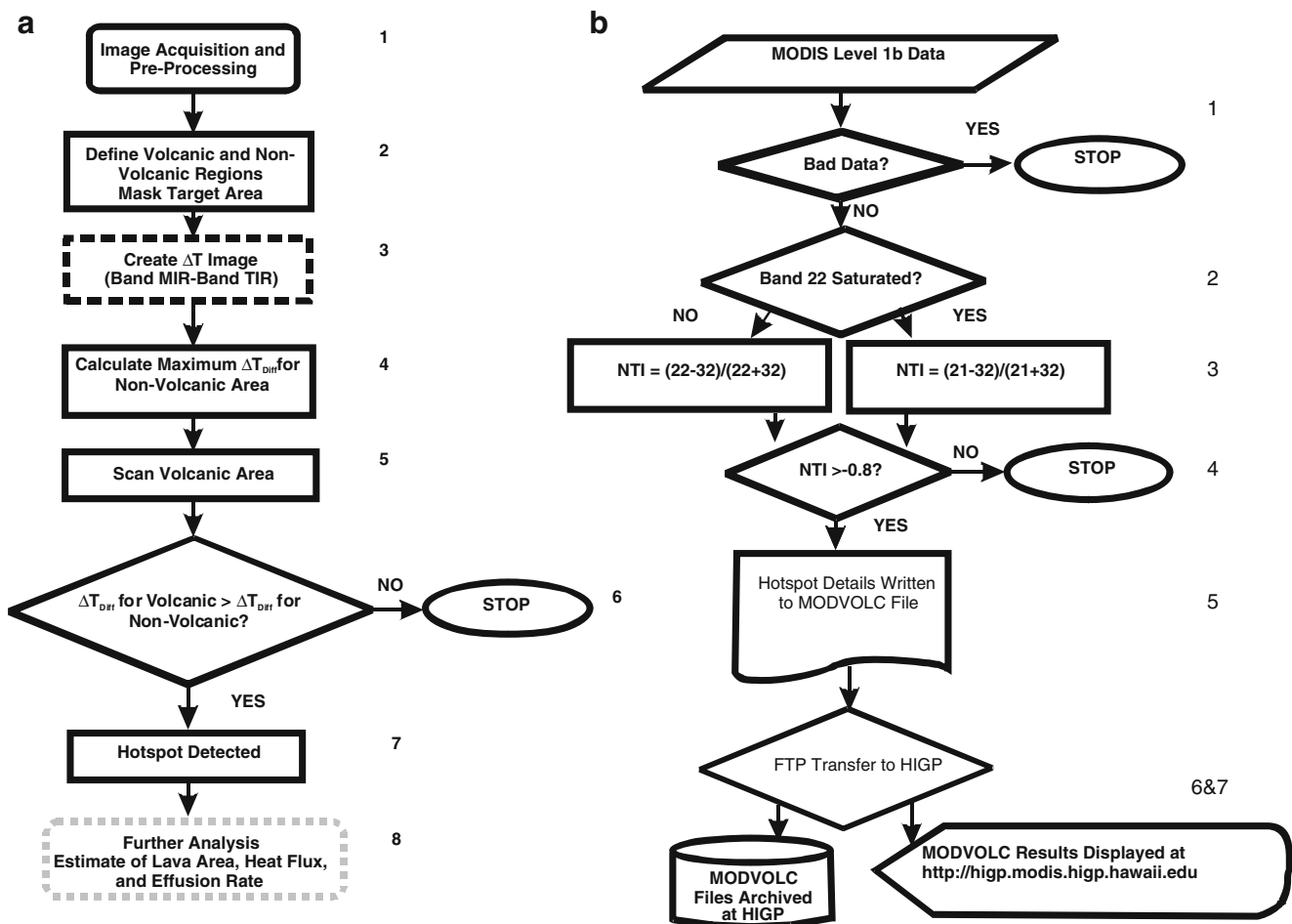


Fig. 3 **a** Flowchart giving operations executed by VAST (adapted from Higgins and Harris 1997). The dashed boxes indicate changes made to VAST for this study, black dashed lines indicate steps that are

no longer incorporated, and gray dashed lines indicated steps that have been added. **b** MODVOLC flowchart giving operations executed by MODVOLC (adapted from Wright et al. 2002)

no longer uses the ΔT image. This adaptation was made because when large high-temperature features are present in a pixel, not only will the MIR temperature become elevated, but also that in the TIR. As a result, ΔT will begin to decline once a hot spot reaches a sufficiently large size (Fig. 1c). In addition, AVHRR's MIR band (band 3) saturates at 50–65°C, capping T_{MIR} and allowing T_{TIR} to gain on the fixed T_{MIR} , thus decreasing ΔT . Hence, the new version of VAST functions in the same way as the original version, but instead uses the pixel-integrated temperature in the TIR rather than ΔT , a modification which also allows application when MIR data is lacking. A user interaction feature was also added allowing pixels not flagged as anomalous (but which need to be considered) to be selected, or falsely identified pixels to be deselected.

Robust Satellite Technique

RST, as applied to volcanic hot spot location by Di Bello et al. (2004) and Pergola et al. (2004), relies on a data archive

of MIR data to create an ALICE (x, y, t). This provides an estimate of how much a pixel brightness temperature diverges from its normal conditions as determined from the data time series. ALICE can be mathematically defined as (Tramutoli 1998):

$$\sigma_V(x, y, t) = [V(x, y, t) - V_{REF}(x, y)] / [\sigma_V(x, y)] \quad (4)$$

$V(x, y, t)$ being the signal measured by the satellite sensor at location (x, y) and time (t), and $V_{REF}(x, y)$ and $\sigma_V(x, y)$ being the reference fields computed for the same location and time.

As shown in Fig. 4, the algorithm executes five steps. In step 1, data are radiometrically calibrated and geolocated, and step 2 determines the reference fields for each pixel. The first step involves calculation of the average and standard deviation of the brightness temperature for each pixel in the time domain. ALICE then quantifies the amount a pixel value varies from its mean, calculated from images acquired during the same time of day, for a month of data (step 3). Step 4 determines whether the ALICE is greater than the

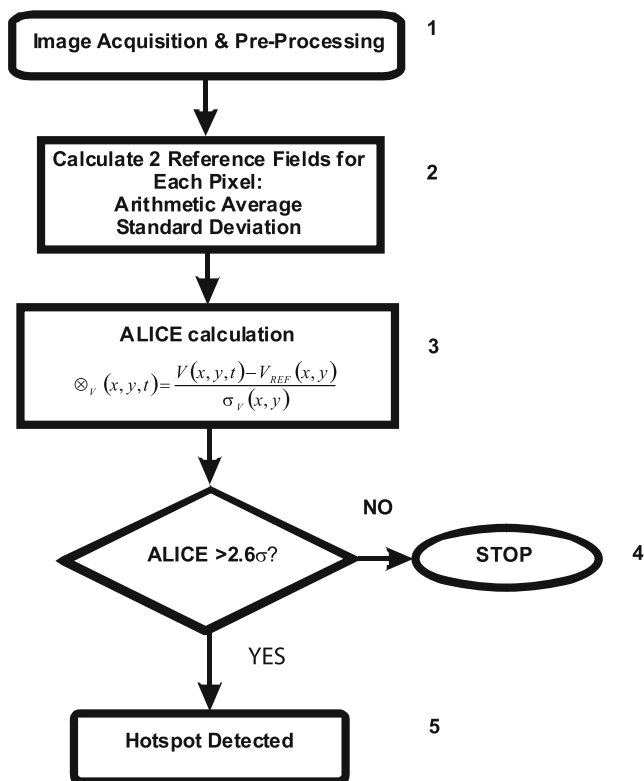


Fig. 4 RST flowchart giving operations executed by RST as described by Pergola et al. (2004)

determined standard deviation. If this case is met, step 5 flags the pixel as thermally anomalous. The standard deviation threshold is designated by the user, where we used a value that is 2.6 standard deviations above the mean, so that a pixel has a 99% probability of being anomalous. This standard deviation works well for the case that it was designed for, sustained lava flows, but modifications might be needed to detect other types of volcanic activity.

Hybrid algorithms

As collated in Fig. 2, a variety of algorithms have been adapted to automatically detect volcanic hot spots. Several of these algorithms do not fall under one of the classification types, but instead incorporate different aspects of each. An example of such a hybrid case is the Okmok algorithm developed at the Alaska Volcano Observatory to track volcanic activity across Alaska, the Aleutians, and Kamchatka (Dean et al. 1998). Okmok incorporates aspects of both the contextual and temporal approaches, detecting thermal anomalies by assessing whether a target pixel is elevated above the mean temperature of a 40×40 pixel sub-image. If this case is met, then the pixel is termed anomalous. The algorithm also incorporates a temporal check which assesses whether the pixel's variation in temperature with time is above that expected from diurnal and seasonal temperature

fluctuations (Dean et al. 1998; Dehn et al. 2000). The MYVOLC algorithm of Hirn et al. (2008) is another hybrid algorithm which combines the fixed threshold MODVOLC algorithm and the contextual VAST algorithm, with MODVOLC2 of Koeppen et al. (2010) combining MODVOLC and RST.

Algorithm assessment: test cases and data

To assess the performance of each algorithm type, we applied VAST, MODVOLC, and RST to the same six test cases. These test cases, as located in Fig. 5 and detailed next, were selected to cover the full range of expected volcanic hot spots from intense, i.e., hot spots that any algorithm must be expected to detect—as during lava flow emission, to weak, i.e., difficult to detect, as during fumarolic activity or persistent open vent degassing and explosive activity.

Etna 2004–2005: sustained lava flows

The 2004–2005 summit eruption of Mt. Etna (Italy, 37.734° N, 15.004° E) began on 7 September 2004 and was characterized by low, but steady, effusion at $2.3\text{--}4.1 \text{ m}^3 \text{ s}^{-1}$ to feed lava flows that extended 2.5 km (Burton et al. 2005). The eruption lasted 6 months ending in March 2005 (Allard et al. 2006). Due to the high surface temperatures at active lava flows, the areas of lava erupted and the persistence of the hot spot, this type of activity should be easiest to detect. As shown in Fig. 6, pixel-integrated temperatures in the MIR and TIR will increase quickly once a sub-pixel lava at temperatures of between 100°C and $1,000^\circ\text{C}$ begins to spread, rapidly becoming elevated above that of the background. However, the presence of an active lava flow can also force pixel-integrated temperatures in the MIR and TIR to levels above the sensor saturation point (Harris et al. 1995). For example, a $2.5 \times 10^5 \text{ m}^2$ lava flow within a 1.0-km^2 pixel will occupy 25% of the pixel. Given flow surface temperatures in the range 100°C to 500°C , this will result in pixel-integrated temperatures in the MIR of between 61°C and 332°C , and between 44°C and 197°C in the TIR. This would saturate AVHRR's MIR band and MODIS' low-gain MIR band, with saturation also occurring in MODIS' high gain MIR band and all TIR bands for the high surface temperature case (Fig. 6a–b). This will result in the capping problem already mentioned for any ΔT -based algorithm.

Stromboli 2007: small-area lava flows

The 2007 effusive eruption of Stromboli (Aeolian Islands, Italy, 38.789° N, 15.213° E) began on 27 February 2007 and continued until 15 March 2007. High effusion rates

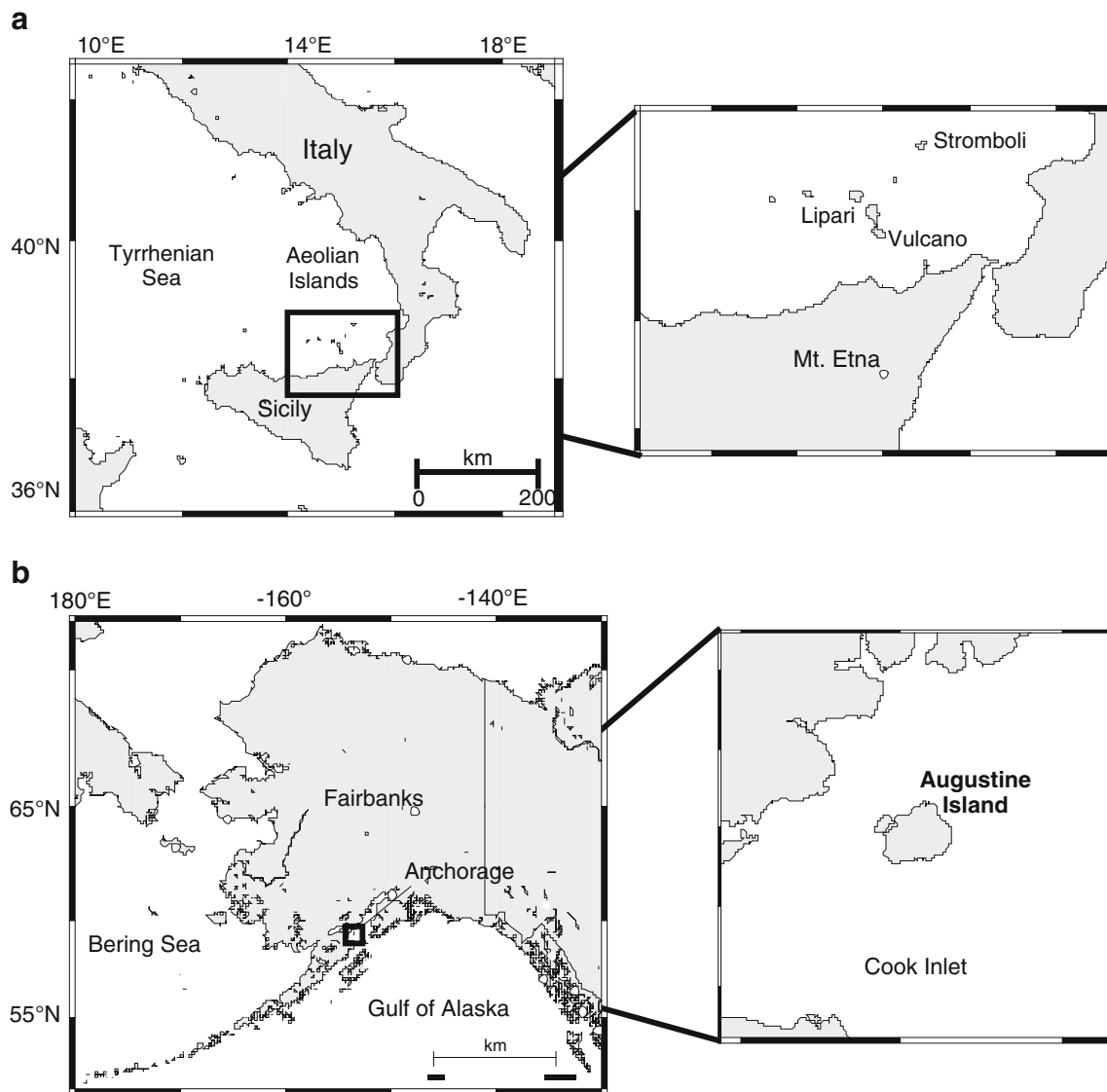


Fig 5 Location map for each study area: **a** Etna, Stromboli, Lipari, and Vulcano, **b** Augustine

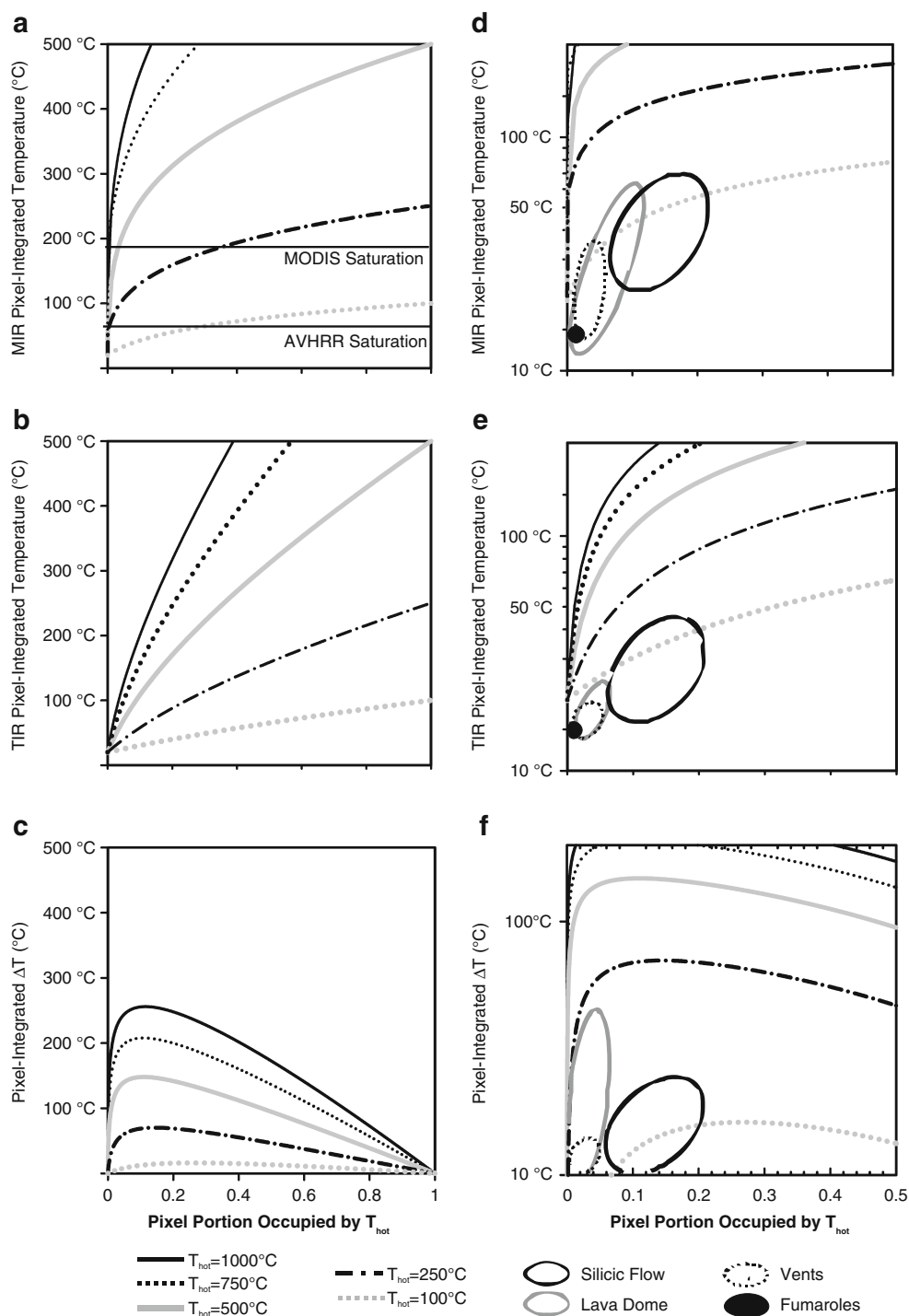
($\sim 20 \text{ m}^3 \text{ s}^{-1}$) initially fed lava flows that extended 1.2 km to the sea (Barberi et al. 2009; Calvari et al. 2010). A high effusion rate phase with a time-average discharge rates of $\sim 8 \text{ m}^3 \text{ s}^{-1}$ characterized the first 6 days of the eruption. Thereafter time-average discharge rates declined to $\sim 3 \text{ m}^3 \text{ s}^{-1}$ feeding flows covering an area of $\sim 1.4 \times 10^5 \text{ m}^2$ during the first day of the eruption, and $0.2 \times 10^5 \text{ m}^2$ by 10 March (Marsella et al. 2009). Although these flows are smaller in area than the Etna flows, this type of effusive activity should still be easily detected using any automated algorithm. In the Stromboli case, the smaller lava flow areas may result in lower pixel-integrated temperatures. For example, a $1 \times 10^5 \text{ m}^2$ lava flow with surface temperatures ranging between 100°C and 500°C in a 1.0 km^2 pixel will occupy 10% percent of the pixel. For such a pixel, pixel-integrated temperatures would be between 42°C and 250°C

in the MIR and between 30°C to 108°C in the TIR. Thus, the feature will be detectable, but the resulting thermal anomalies will not be quite as intense, and pixel-integrated temperatures may be below saturation levels in some MIR and TIR cases (Fig. 6).

Stromboli 2006: open vent degassing and intermittent strombolian activity

Stromboli's normal activity is characterized by persistent open vent degassing from several hot vents distributed across three main craters (Harris and Ripepe 2007b). These vents also feed short-lived, small volume strombolian eruptions that emit hot scoria and bombs around nine times per hour (Patrick et al. 2007). A year's worth of satellite data for Stromboli was analyzed to determine which, if any,

Fig. 6 Change in pixel-integrated temperature with percent coverage by a hot source for mixed pixels composed of two components for the MIR (**a, d**), TIR (**b, e**) and the difference between the two, i.e., MIR minus TIR (**c, f**). Plots are given for a range of hot spot temperatures with a T_{amb} of 20°C. The saturation temperatures of AVHRR and MODIS MIR bands are given using horizontal lines



algorithm can most effectively detect hot spots associated with such activity. The heat sources presented by the open vents should present a detectable hot spot, if enough vents are active and/or vent temperature sufficiently high, as during times of high magma levels (Harris and Stevenson 1997). Given range of scenario's ranging from one to ten 4-m-wide vents at 950°C, such a source should lead pixel-integrated temperatures of between 20°C and 36°C in the MIR, compared with a background of 20°C. However, such

a source will not elevate pixel-integrated temperatures in the TIR (Fig. 6b). Therefore, although there may be detection problems if the source is small, we are unlikely to encounter saturation problems and ΔT can present an effective means to detect such activity once an anomaly forms in the MIR (Fig. 6c). Note here that it is the heat source due to the open vents that we are detecting, not the explosions themselves. At Stromboli, the thermal signature from the explosions will be difficult to detect and isolate for

several reasons. First, they will have to occur at the time of, or shortly before, the satellite is overhead collecting data, so that the chances of coinciding with a 10–20-s long event that occurs nine times an hour is small. Second, the deposits will cool very rapidly and are not particularly extensive, so they will not give a strong hot spot. Third, it is likely that the hot vents themselves will dominate the signal (Piscopo 2010).

Augustine 2006: lava dome growth and collapse

Augustine volcano (Alaska, USA, 59.326° N, 153. 435° W) is a small dome complex located in the lower Cook Inlet of Alaska (Fig. 5). Activity is characterized by explosions usually followed by the emplacement of an andesite–dacite summit lava dome (Miller et al. 1998; Power et al. 2006). For this study, the occurrence of thermal anomalies during the 2006 eruption were examined. For this eruption, Power et al. (2006) define an explosive phase between 11 and 28 January, followed by a period of continuous activity characterized by frequent small ash-emitting explosions and dome collapses producing pyroclastic flows until 2 February. This was followed by an effusive phase during which the dome experienced exogenous and endogenous dome growth and began to refill the crater until 31 March. Because there was a variety of activity styles during this eruption, the performance of the different algorithms will also likely vary during the eruption. During the explosive phase, the occurrence of ash plumes will affect whether the satellite sensors are able to detect thermal anomalies, their presence masking the hot source from the satellite view. Also, during the latter portion of the effusive phase, the algorithm may not be capable of detecting a cool dome carapace, where temperatures on active silicic lava flows have been shown to approach ambient (Harris et al. 2002a; Harris et al. 2004). However, surface temperatures can be more highly elevated during phases of exogenous growth (Oppenheimer et al. 1993; Wooster and Kaneko 1998). Thus, as plotted in Fig. 6, a wide range of sizes and surface temperatures can characterize an active silicic lava dome or flow. Surface temperatures obtained for Unzen by Wooster et al. (2000) ranged between 400 and 500°C over areas of between 60 and 1,600 m². Such a hot spot would occupy between 0.1% and 0.006% of a 1-km² pixel. The resulting MIR pixel-integrated temperatures would fall between 22°C and 52°C (Fig. 6a–d), with TIR pixel-integrated temperatures ranging between 20°C and 22°C (Fig. 6b–e). The resulting ΔT would range between 0°C and 22°C (Fig. 6c). Therefore emplacement of such a lava dome would most easily be detected in the MIR, but not in the TIR, with the sub-saturation levels meaning that ΔT approaches can be implemented to detect such activity (Fig. 6c–f).

Vulcano 2006: fumarolic activity

Vulcano (38.4° N, 14.967° E) is the southernmost of the Aeolian Islands (Fig. 5) and hosts an active fumarole field covering an area of ~415,000 m² (Chiodini et al. 2005). Individual vent areas are, however, small (typically being millimeters to centimeters in diameter) and may only occupy ~5% of the field area (Harris et al. 2009). Vent temperatures are also low, ranging between 80°C and 430°C (mean=183°C, no.=591) in June 2006 (Harris, unpublished data). Because the temperatures of the fumaroles are low and vent areas small, the thermal anomaly should be extremely difficult to detect in a 1-km pixel. For example, the lower zone of Vulcano's fumarole field had a total vent area of 9 m² with a maximum temperature of 235°C and a mean of 110°C in July 2005 (Harris et al. 2009). Such a heat source against an ambient background of 20°C would not be detectable, although the 1,600 m² of higher temperature emission in the mid-zone (Harris et al. 2009), where temperatures had a mean of 177°C (no.=220) in June 2006, may be detectable.

Lipari 2006: no volcanic hot spot

Lipari Island (38.48° N, 14.95° E) is the largest of the Aeolian Islands and was active from 223 ka to historic times, the most recent eruption resulting in both explosive and effusive eruption of rhyolite in 580 AD (Cortese et al. 1986; Crisci et al. 1991). Thus, Lipari was chosen as a null test case to examine how often the algorithms detected a volcanic hot spot at a potentially active, but otherwise cold and inactive, volcano.

Algorithm performance tests: methodology

Each of the algorithms were run and compared with results obtained from manual image inspection using the two most commonly used data types: AVHRR and MODIS (as listed in Table 3). The manual inspection served as the benchmark, where every image used for the tests was visually inspected for image quality (e.g., the presence of bad data and large zenith angles), cloud conditions and presence of real hot spots associated with volcanic activity. If a hot spot was apparent the number of anomalous pixels was recorded, along with their locations and radiances (or temperatures) in both the MIR and TIR. A pixel was considered to be anomalous if the pixel-integrated temperature was elevated by $\geq 5^\circ\text{C}$ above its surrounding (non-anomalous) pixels. In order to elevate a 1-km pixel 5°C above a background temperature of 20°C, the sub-pixel hot spot must be greater than $2.1 \times 10^4 \text{ m}^2$ (for a 100°C source) or 5,000 m² (for a 500°C source) (Fig. 6).

Each algorithm was then run on the same data sets and the output compared with the handpicked results. AVHRR data were not used for testing of the RST algorithm. Execution of RST requires precise geometric rectification so that exactly the same pixel is considered for the statistical tests. Large errors associated with the geolocation of AVHRR images (up to 20 km) thus meant that the algorithm was run using MODIS data whose geolocation is much more accurate (<1 km).

The algorithms were first evaluated in terms of how many of the images in which hot spots had been manually confirmed were flagged. These results were used to test if the algorithm was capable of at least confirming the presence of a hot spot in any given image. Next, the algorithm output were examined to see how many of the anomalous pixels identified in each image by manual inspection were detected by the algorithm. This was used to assess how well each algorithm can detect and locate all of the anomalous pixels present in any one image. One way of assessing the performance of the algorithm in identifying hot spot containing images and anomalous pixels is through scatter plots of the number of manually identified cases versus algorithm identified cases (e.g., Fig. 7). If the plots fall on the 45° line of Fig. 7, then the performance is perfect, plots falling above this line identify the incidence of false detection, plots falling below the line identify cases where the hot spot image (or pixel) is not detected.

Algorithm performance tests: results

Etna 2004–2005: sustained lava flows

Etna's 2004–2005 eruption lasted 207 days, during which time 670 nighttime AVHRR images were acquired. Of these, 249 were cloud free and of suitable quality for hot spot detection. Hot spots were manually identified in 132 images, within which 580 pixels were identified as anomalous. Over the same period, 697 nighttime MODIS images were acquired, of which 262 were of suitable quality for detection. A total of 234 images had thermal anomalies, with a total of 1,751 anomalous pixels.

VAST suggested that hot spots were present in 146 of the AVHRR images acquired during the eruption period. This included all 132 of the AVHRR images on which a hot spot had been manually identified. However, 14 (10%) of the flagged cases were false detections. A total of 832 pixels were flagged by VAST, of which 315 (46%) were false detections. VAST also missed 63 pixels identified by manual analysis, so that 10% of the anomalous pixels were not detected (Fig. 7a).

Of the 225 MODIS images that contained hot spots, MODVOLC correctly identified hot spots in 212 (94%), thereby missing 13. The scarcity of points above the 45° line

in Fig. 7c indicates a very low incidence of false detections with MODVOLC, there being just one. Within the 212 hot spot images flagged by MODVOLC, 1084 anomalous pixels were detected. The algorithm thus missed 687 (38%) of the anomalous pixels. Of the 1,084 pixels detected, just one proved to be a false detection (Fig. 7). MODVOLC's performance when run using AVHRR data was not as good. Although no false detections were returned, hot spots were found in just 57 (43%) of the 132 hot spot containing AVHRR images (Fig. 7b). Also, only 166 (27%) of the 580 anomalous pixels were found, with four (2%) being false detections (Fig. 7b).

RST identified hot spots in 160 (71%) of the 225 hot spot containing MODIS images, with one false positives (0.4%). Of the 1,800 anomalous pixels identified manually, 795 (45%) were detected by RST meaning that 1,005 (55%) thermally anomalous pixels were missed. Of the pixels detected, 13 (1.6%) were false detections (Fig. 7d).

Stromboli 2007: small-area lava flow

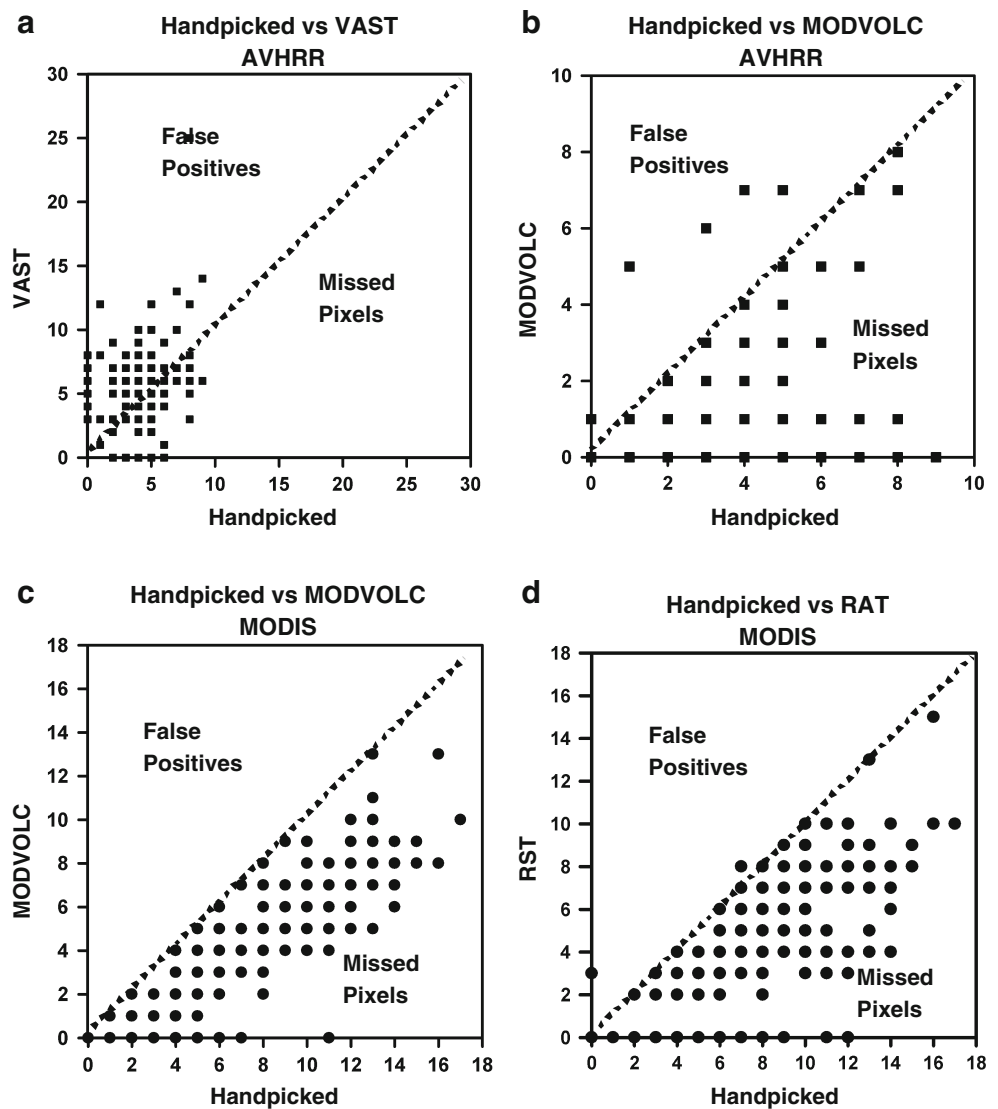
During Stromboli's 2007 effusive eruption, a total of 181 AVHRR and 118 MODIS images were acquired over the 17-day eruptive period. Of these, 30 AVHRR and 26 MODIS images were of sufficient quality for hot spot detection. Manual inspection revealed that 18 of the AVHRR images, and 21 of the MODIS images, contained hot spots. Within these images, 78 anomalous AVHRR pixels and 160 anomalous MODIS pixels were identified.

VAST identified flagged all 18 hot spot containing images, with no false detections (Fig. 8). Across these images, VAST flagged 87 pixels as anomalous (Fig. 8), of which nine were false detections, and missed 40 pixels (51%). Therefore VAST identified 100% of the images in which hot spots had been identified, but 11% of the flagged pixels were false detections (Fig. 8a).

Of the 21 cloud-free MODIS images that contained hot spots, MODVOLC identified hot spots in 18 (86%), flagging a total of 69 pixels (43%). The occurrence of false detection was zero in terms of both images and pixels (Fig. 8c). When MODVOLC was applied to the 18 hot spot containing AVHRR images, although there were no false detections, only nine images (50%) were determined to contain hot spots, across which 12 pixels were identified. Therefore MODVOLC missed 50% of the AVHRR images in which hot spots had been manually identified, and only 14% of the anomalous pixels were found (Fig. 8b).

RST identified hot spots in just eight of the 21 MODIS images that contained hot spots, locating a total of 42 anomalous pixels (Fig. 8d). Therefore, although RST produced no false detections it identified just 38% of the MODIS images containing hot spots and 26% of the anomalous pixels.

Fig. 7 Scatter plots of hand-picked versus algorithm detected anomalous pixels for the Etna 2004–2005 eruption. Plots show the number of pixels detected by VAST, MODVOLC, and RST compared with those determined manually. Points that lie above the dotted line represent false positives and points that lie below represent pixels that were not detected by the algorithms



Stromboli 2006: open vent degassing and intermittent strombolian activity

During 2006, a total of 899 AVHRR and 1707 MODIS nighttime images were acquired over Stromboli. Of these, 468 AVHRR and 386 MODIS images were cloud free. Manual inspection showed that a total of 46 AVHRR images contained hot spots, within which 65 anomalous pixels were evident. Hot spots were identified in 52 of the MODIS images, across which 69 anomalous pixels were identified.

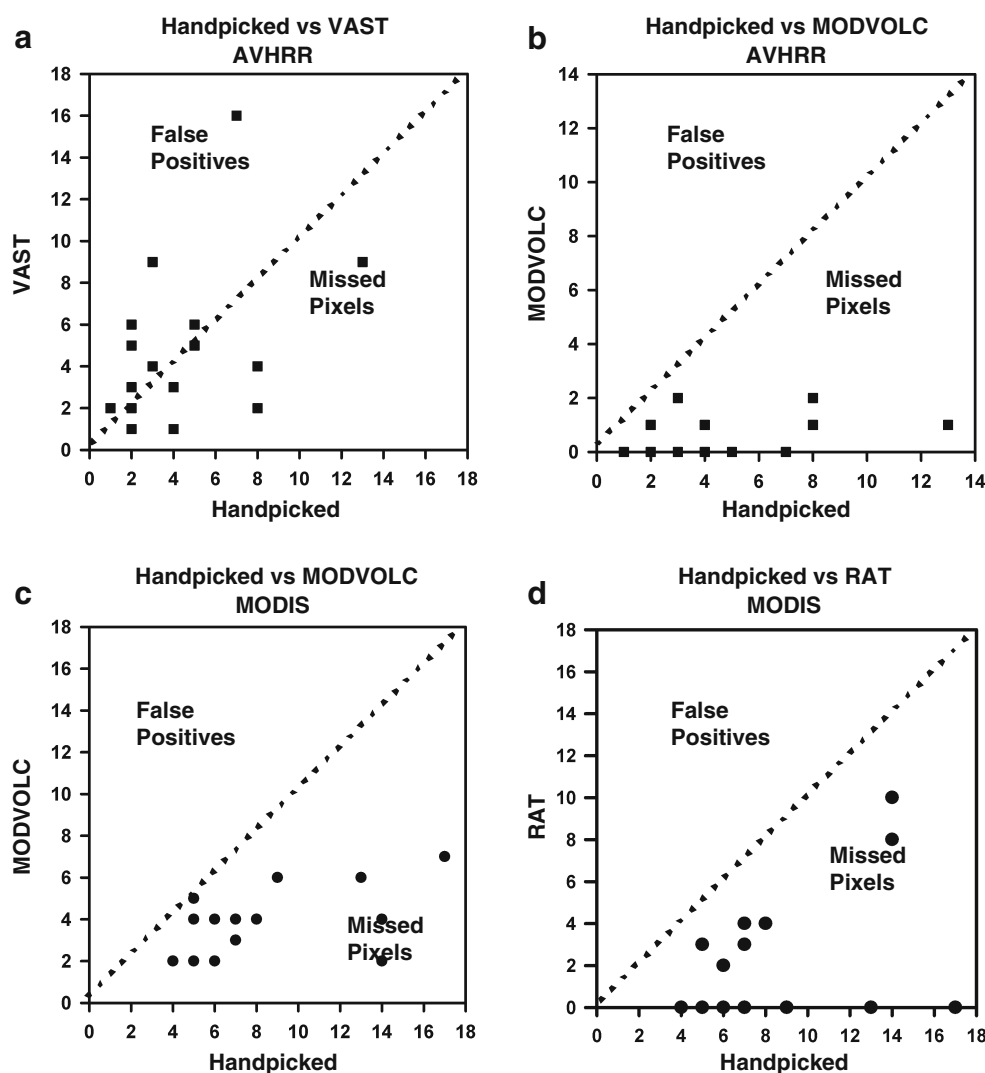
VAST suggested the presence of hot spots in 66 of the AVHRR images collected. However, 46 were false detections and 23 of the images in which hot spots had been found were missed. Thus, while 68% of the images flagged by VAST were false detections, 50% of images in which hot spots were apparent were missed. A total of 97 pixels were located by VAST, of which 66 (68%) were false detections (Fig. 9a). VAST also missed 33 (50%) of the anomalous pixels. The overall occurrence of false detections, taking

into account all 899 AVHRR images run through the algorithm, was 12%.

MODVOLC flagged hot spots in eight MODIS images, across which a total of 12 pixels were flagged as anomalous. Therefore, MODVOLC identified 15% of the anomalous images and 17% of the anomalous pixels (Fig. 9c). Hence, MODVOLC missed hot spots in 86% of the images, with one detection being false (Fig. 9a). When MODVOLC was run using AVHRR data, just two images were flagged, with one anomalous pixel being found in each (Fig. 9b). Therefore, although the occurrence of false positives was zero, MODVOLC missed hot spots in 97% of the AVHRR images.

RST identified hot spots in 23 of the MODIS images, across which 34 anomalous pixels were flagged. RST thus detected 33% of the hot spot images, and 32% of the anomalous pixels. Six images were also falsely identified as containing hot spots, so that 17% of the detections were false (Fig. 9d).

Fig. 8 Scatter plots of hand-picked versus algorithm detected anomalous pixels for the Stromboli 2007 eruption. Points that lie above the dotted line represent false positives and points that lie below represent pixels that were not detected by the algorithms



Augustine 2006: lava dome growth and collapse

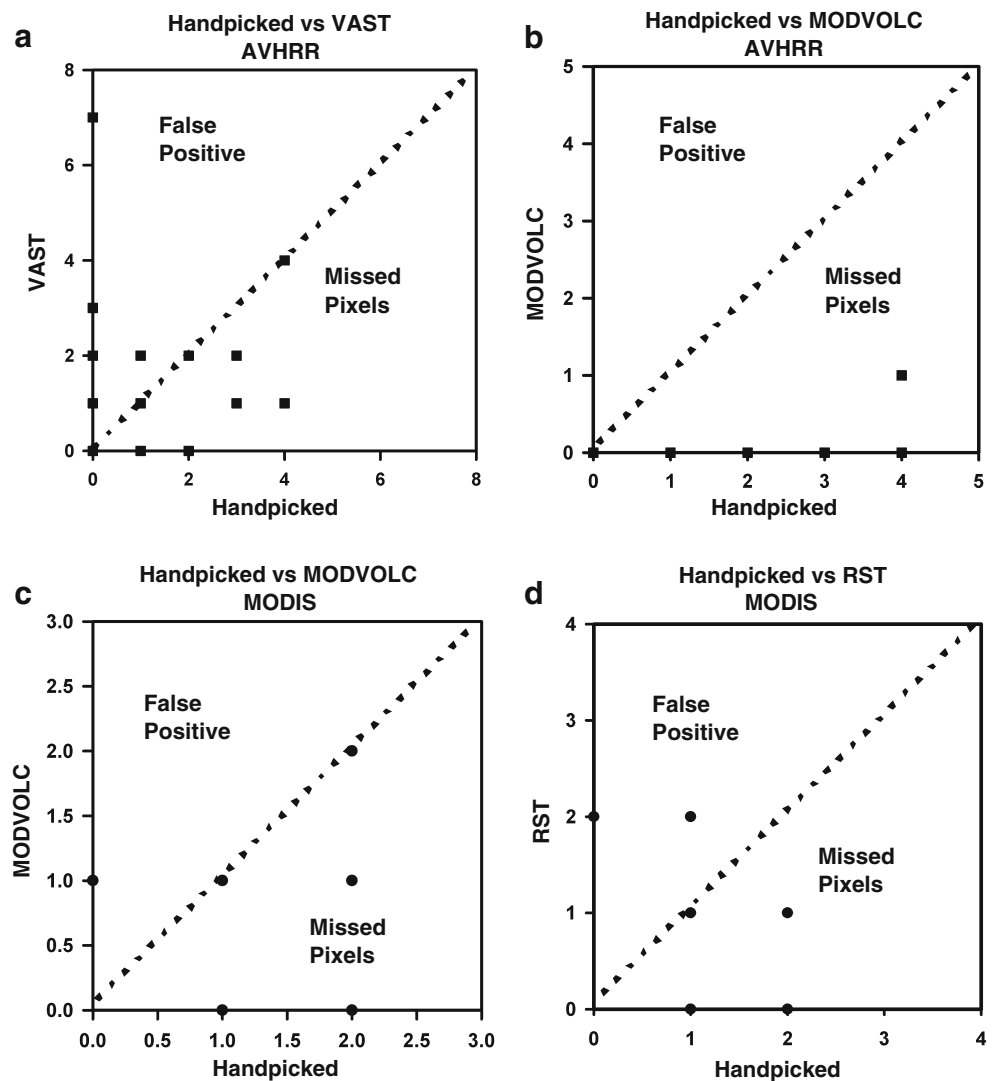
During the 48-day-long 2006 eruption, 1,230 AVHRR nighttime images were collected, of which 403 were cloud free and 289 contained hot spots. Within these images, a total of 2,377 pixels were found to be anomalous. During the same period, 528 MODIS nighttime images were collected, of which 149 were of high-quality and cloud free. Manual inspection revealed that 68 images contained hot spots, within which a total of 384 anomalous pixels were identified.

VAST identified 147 AVHRR images as containing hot spots, 17 of which were false detections. Therefore, VAST correctly identified 45% of the hot spot images, returning false detections for 12%. A total of 520 pixels were flagged by VAST, with 82 being false detections. Therefore VAST identified 18% of the anomalous pixels during the eruption period, with 15% of the detections being false (Fig. 10a). When considering all 1,230 AVHRR images run through the algorithm, the false detection incidence was 6%.

MODVOLC detected thermal anomalies in 38 (60%) of the 64 MODIS images containing hot spots, with no false detections. In all, 157 pixels were located, with no false detections (Fig. 10c). Therefore MODVOLC correctly identified 41% of the anomalous pixels, with 0% of the pixels flagged being false. When considering all 528 MODIS images run through the algorithm, the false detection rate for MODVOLC was less than 0.1%. MODVOLC identified hot spots in 79 (27%) of the AVHRR images, with one of them being a false detection (Fig. 10b). Within these, MODVOLC located 293 (12%) of the anomalous pixels, of which two (0.6%) were false detections (Fig. 10b).

RST identified 52 MODIS images as anomalous, of which one was a false detection. RST therefore flagged 82% of the hot spot images across which 310 pixels were identified as anomalous, 31 (9%) of which are false detections (Fig. 10d). RST's total occurrence of false detections when run for the entire Augustine data set was 2%.

Fig. 9 Scatter plots of hand-picked versus algorithm detected anomalous pixels for Stromboli 2006. Points that lie above the dotted line represent false positives and points that lie below represent pixels that were not detected by the algorithms



Vulcano 2006: fumarolic activity

During 2006, 324 AVHRR images and 460 MODIS nighttime images were collected over Vulcano. Visual inspection determined that no thermal anomalies were present. Although MODVOLC and RST did not detect any hot spots when run with MODIS or AVHRR data, VAST identified 58 AVHRR images as anomalous, within which 106 pixels were flagged (Fig. 11a). Therefore, VAST had an 18% occurrence rate of false detections (number of false detections divided by the total number of cloud-free AVHRR images).

Lipari 2006: null case

During 2006, 460 MODIS and 319 AVHRR images of Lipari were acquired and run through VAST, MODVOLC and RST. Through visual inspection one of the images were found to contain a hot spot, likely due to a fire. MODVOLC did not identify any image as anomalous. VAST falsely identified a total of 27 images as containing

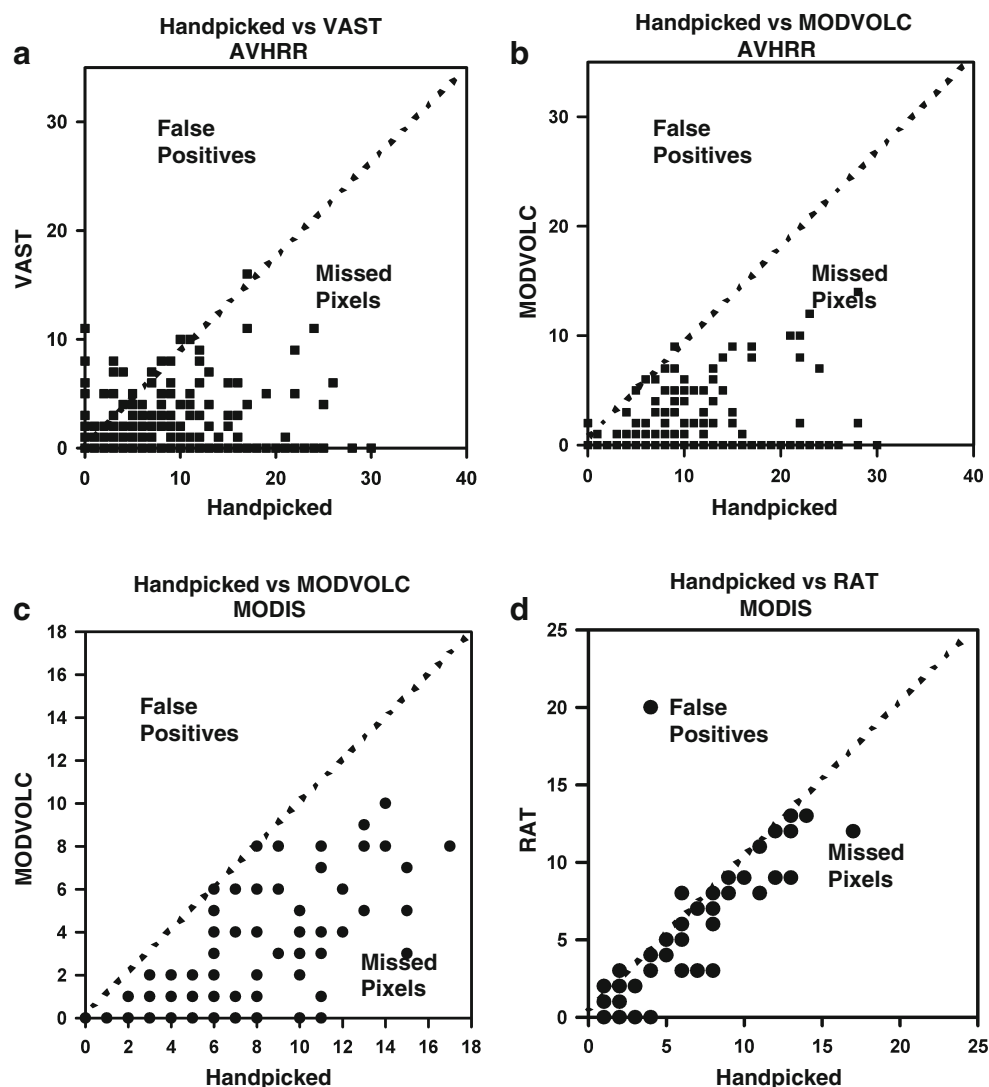
hot spots, across which 39 pixels were classified as anomalous (Fig. 11c). Therefore the incidence of false detection was 8%. RST produced two false detections, within which three pixels were identified, resulting in a 0.4% false detection incidence (Fig. 11b).

Discussion I: algorithm performance

Volcanic Anomaly Software

For active lava flows, VAST offers excellent detection performance, correctly identified hot spots in 98% of the images that contained hot spots during Etna's 2004–2005 eruption, and 100% of the images during Stromboli's 2007 effusive eruption (Tables 4 and 5). However, this comes at the expense of a false detection incidence of up to 10%. The success rate in terms of correctly identified pixels was slightly lower, being 89% and 49% for the two cases, respectively (Tables 4 and 5). Therefore VAST detected at

Fig. 10 Scatter plots of hand-picked versus algorithm detected anomalous pixels for the Augustine 2006 eruption. Points that lie above the dotted line represent false positives and points that lie below represent pixels that were not detected by the algorithms



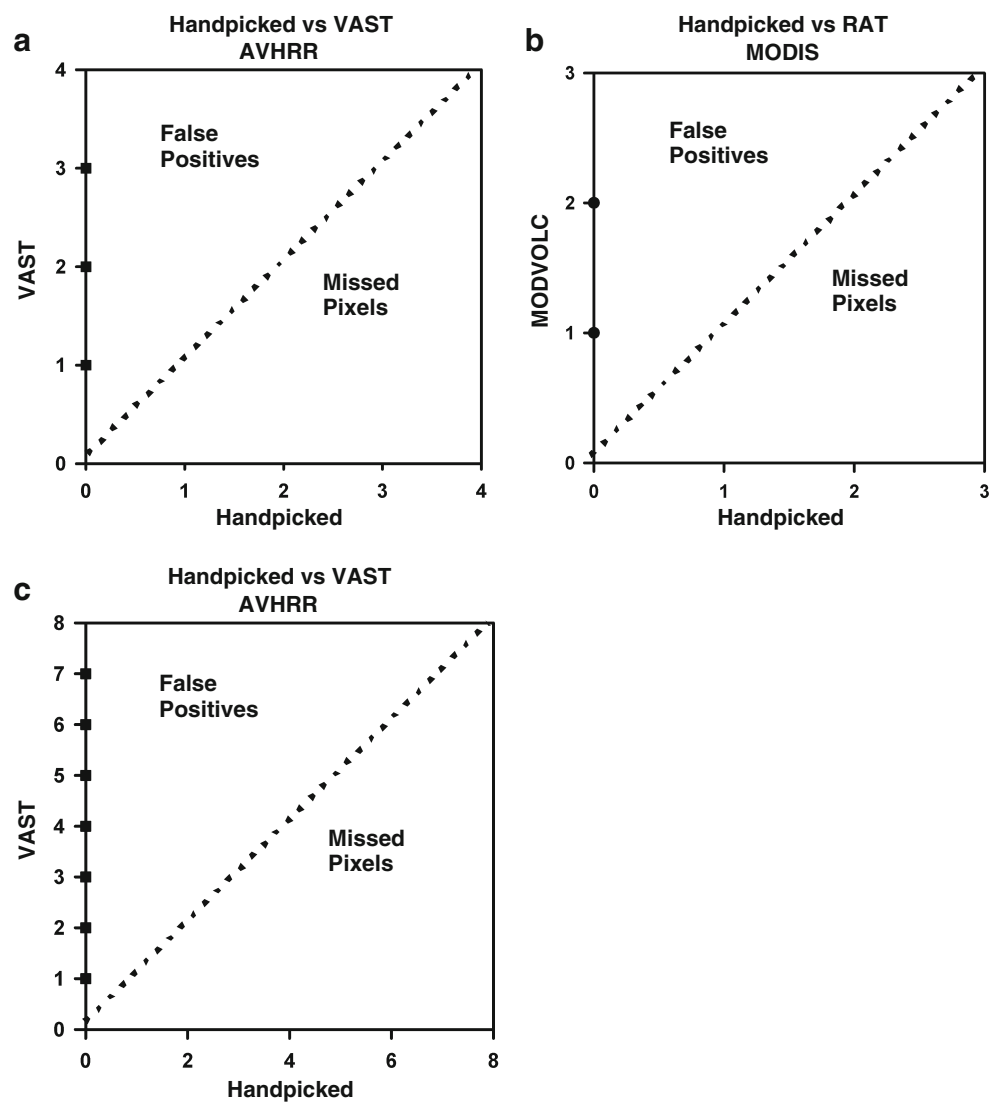
least one pixel in all of the anomalous images, but did not detect all of the pixels in some of the images (Fig. 12a). The high accuracy is expected because VAST was developed to detect high-temperature lava flows, which are apparent in the TIR bands (Fig. 6e).

The false detections produced during the Etna eruption occurred along the Sicilian coast and on the lower flanks of the volcano (Fig. 12c–d). Coastal false positives occur when the temperature difference between the land and the sea are higher than the calculated temperature difference within the defined non-volcanic region. The defined volcanic region for Etna contains a small portion of coastline, so that the occurrence of these false positives can easily be alleviated by redefining the volcanic region and ensuring that coastal zones are not within the region. For example, the 25×25 box for the AVHRR image acquired on 28 November 2004 at 04:04 UTC included ten coastal pixels and a ΔT_{diff} threshold of 2°C . As a result one false positive was generated at the coast (Fig. 12c).

However, decreasing the volcanic region to 20×20 pixels excludes the coastal pixels and results in no false detections. These false detections could also be eliminated if ΔT images were used instead of the TIR image, as implemented in the original algorithm. We repeat, though, that changes to the algorithm needed to be made because of saturation and calibration problems associated with the Band 3 AVHRR data considered here. False positives also frequently occurred at lower elevations on the volcano. At these locations, the lower elevation pixels are warmer than the adjacent higher, cooler (sometimes snow covered) summit region and are therefore flagged as anomalous (Fig. 12e). Again these false positives would be minimized if the ΔT image was used.

VAST did not perform as well in detecting hot spots associated with normal strombolian activity at Stromboli, with only 50% of the hot spot containing images being identified and a 44% false detection incidence (Table 6). In terms of pixels, VAST only located 49% of the anomalous pixels,

Fig. 11 Scatter plots of hand-picked versus algorithm detected anomalous pixels for Vulcano and Lipari 2006. Points that lie above the dotted line represent false positives and points that lie below represent pixels that were not detected by the algorithms. Note that only false positives were detected by VAST and RST during 2006



with 68% of all located pixels being false (Table 6). The decrease in performance when applied to this case is to be expected considering the principle on which VAST operates. Our modified version of VAST uses AVHRR band 4 temperatures (T_{TIR}) to detect thermal anomalies. As already shown, given the small area of the vents at Stromboli they are best detected in the MIR (band 3) rather than the TIR (band 4) (Fig. 6d–e). Again, false detections occurred along the coastline of the island, with the sea pixels being warmer than their neighboring island pixels by up to 6°C. In addition, the temperature differences across the pixels composing the non-volcanic zone was not that great, it being represented by a homogenous sea surface across which temperatures underwent little (<2°C) variation. Thus, the threshold was often too low when applied to the island, so that when the island was hotter than the surrounding ocean, we simply detect the presence of the island. As for the Etna case, the occurrence of these false detections could be rectified if a ΔT image was used instead of the TIR image.

VAST identified 45% of the anomalous images at Augustine, with false positives occurring 17% of the time. The number of pixels detected was low with 18% of the anomalous pixels being identified, although false detections were only 3% (Table 7). As at Stromboli, one of the reasons that VAST's performance was poor was because lava dome growth is not as apparent in the TIR (AVHRR's band 4) as it is in the MIR (AVHRR's band 3). As already discussed, the low temperatures typically encountered at a lava dome surface can result in a weak or low intensity TIR anomaly (Fig. 6e). Therefore the temperature differences in the TIR (band 4) are not as prominent as in the MIR (band 3).

Although false positives did not occur frequently, when they did occur they again occurred on the coastline of Augustine Island. For Vulcano and Lipari, VAST only generated false detections (Tables 8 and 9). All occurred at the coastlines. As at Stromboli, this problem is difficult to correct when dealing with small volcanic islands where the fairly constant temperature of the sea surface used to set the

Table 4 Number of correct, missed, and false detections for Etna 2004–2005 effusive eruption

Algorithm	Manual		VAST	MODVOLC		RST
	AVHRR no. (%)	MODIS no. (%)	AVHRR no. (%)	AVHRR no. (%)	MODIS no. (%)	MODIS no. (%)
Images	132	225				
Correct			129 (98)	57 (43)	212 (94)	160 (71)
Missed			3 (2)	75 (57)	13 (6)	65 (29)
False			17 (10)	0 (0)	0 (0)	1 (0.4)
Pixels	580	1,800				
Correct			517 (89)	162 (28)	1,065 (59)	795 (45)
Misses			63 (11)	418 (73)	687 (38)	1,005 (55)
False			315 (38)	4 (2)	0 (0)	13 (1.6)

threshold can cause the T_{diff} threshold to be smaller than the temperature difference between the island and surrounding sea pixels. These problems could be rectified by reverting to the original version of VAST that uses a ΔT image. However, problems due to differential emissivity (ε) between the MIR and TIR over some volcanic surfaces, which can cause ΔT to be up to 8°C (see Figure 7 of Higgins and Harris 1997), may persist if brightness temperature are not corrected for ε before calculation of ΔT .

MODVOLC

MODVOLC also performed when applied to detect lava flows at Etna and Stromboli using MODIS data. The large size and high temperatures of lava flow surfaces make them easily detected by the NTI technique even with a high (globally applicable) threshold. MODVOLC flagged 95% and 80% of the anomalous images at Etna and Stromboli, respectively (Tables 4 and 5), with the occurrence of false detections being 1% for Etna and 0% for Stromboli. Although MODVOLC was effective in detecting thermal anomalies, it was not as effective in identifying all of the

anomalous pixels in the images (Fig. 13a–j), with just 60% of all anomalous pixels being detected at Etna and 43% at Stromboli. Examples of how MODVOLC detected the pixels with the highest radiances, but missed some of less radiant pixels are seen in Fig. 13a–j. This is to be expected because MODVOLC implements a high NTI in order to eliminate the occurrence of false positives in a global search, and it was only ever MODVOLC's intent to detect the incidence of a hot spot, rather than to serve as a tool to find all anomalous pixels. In order to detect all anomalous pixels the MODVOLC threshold would need to be lowered to a locally appropriate value, as done by Kervyn et al. (2008) as part of MODLEN. However, this would also increase likelihood of false detections.

False detections rarely occurred and in all cases resulted from geolocation errors with the anomaly not actually being located at the volcano of interest. This does not occur frequently and is a result of the algorithm running at the NASA DAAC where mis-registration of the geolocation files can occur, so that the error is not actually associated with the algorithm, but with the NASA-based pre-processing. Because the processing occurs off site this is

Table 5 Number of correct, missed, and false detections for Stromboli 2007 effusive eruption

Algorithm	Manual		VAST	MODVOLC		RST
	AVHRR no. (%)	MODIS no. (%)	AVHRR no. (%)	AVHRR no. (%)	MODIS no. (%)	MODIS no. (%)
Images	18	21				
Correct			18 (100)	9 (50)	18 (86)	8 (38)
Missed			0 (0)	9 (50)	3 (14)	13 (62)
False			0 (0)	0 (0)	0	0 (0)
Pixels	78	160				
Correct			38 (49)	12 (3)	69 (43)	42 (26)
Missed			40 (51)	66 (97)	91 (57)	118 (74)
False			9 (11)	0 (0)	0 (0)	0 (0)

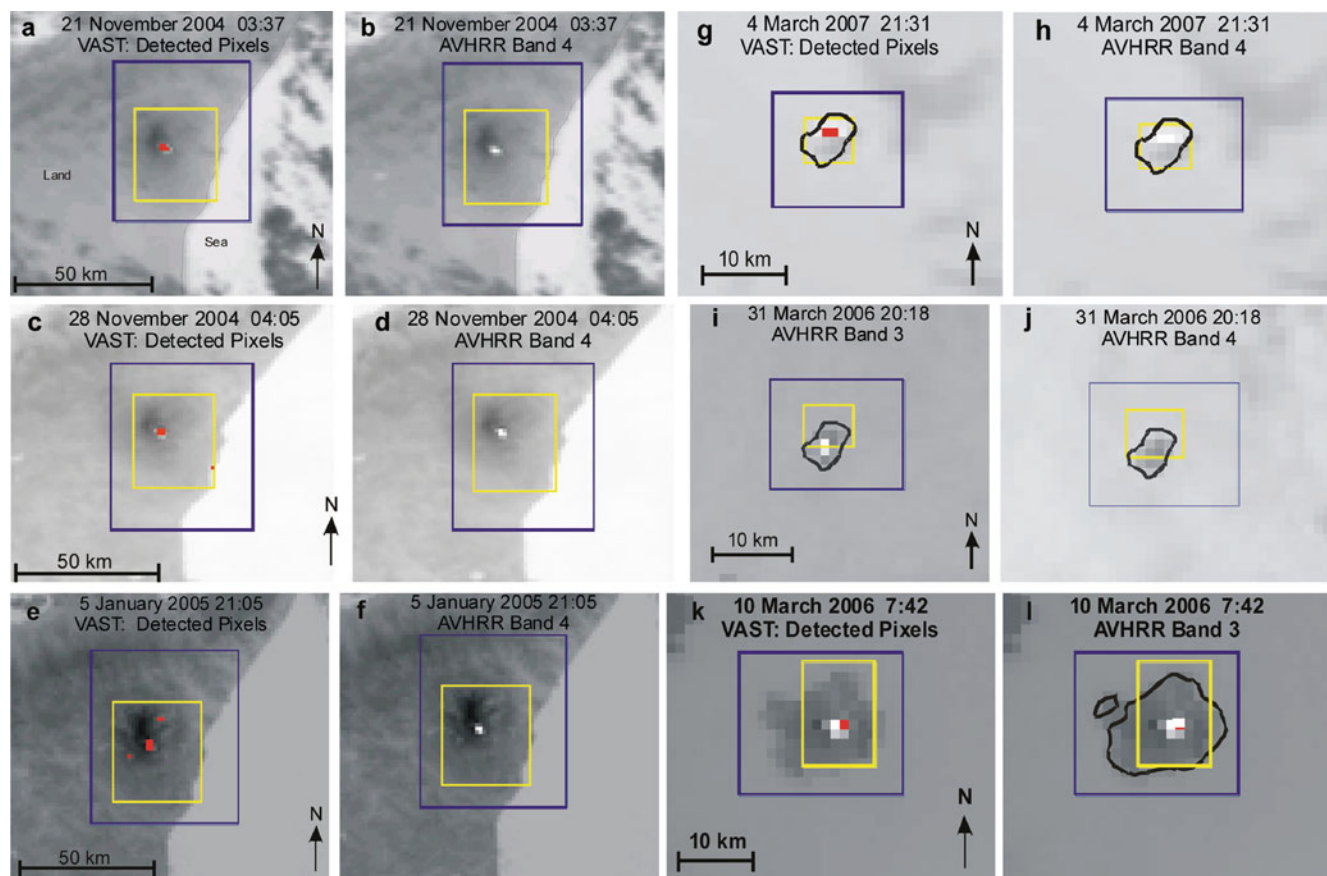


Fig. 12 Examples of volcano hot spot images with pixels flagged as anomalous by VAST. (**a, b**) Good result at Etna: five out of the seven anomalous pixels are flagged. (**c, e**) Bad results at Etna: Examples of typical false positives that occur on the coast of Sicily (**c**) and those that occur and the warmer flanks of Etna (**e**). (**g, h**) Good result at

Stromboli: two out of the four anomalous pixels are flagged. (**i, j**) Example of thermal anomaly missed at Stromboli - the anomaly is apparent in the MIR, but not large or hot enough to raise the pixelintegrated temperature in the TIR. (**k, l**) Good result at Augustine: one out of the five anomalous pixels are flagged

difficult to rectify, but if the algorithm was run onsite it could be easily rectified by assuring that only images over the regions of interest are run.

MODVOLC did not perform as well when using AVHRR data. Only 43% of the lava flow hot spot images

for Etna and 50% of those for Stromboli were flagged. The reason for this is that the algorithm does not function properly in cases where the AVHRR MIR band (band 3) saturates. Given that the MIR band is extremely sensitive to sub-pixel hot spots at lava temperatures, and saturates at a

Table 6 Number of correct, missed, and false detections at Stromboli during 2006

Algorithm	Manual		VAST	MODVOLC		RST
	AVHRR no. (%)	MODIS no. (%)	AVHRR no. (%)	AVHRR no. (%)	MODIS no. (%)	MODIS no. (%)
Images	46	52				
Correct			12 (26)	2 (4)	8 (15)	23 (44)
Missed			34 (74)	44 (96)	44 (85)	52 (56)
False			43 (65)	0 (0)	1 (12)	6 (21)
Pixels	65	69				
Correct			17 (26)	22 (3)	12 (16)	34 (49)
Missed			48 (74)	63 (97)	58(84)	45 (51)
False			66 (68)	0 (0)	1 (8)	10 (23)

Table 7 Number of correct, missed, and false detections during the Augustine 2006 eruption

Algorithm	Manual		VAST	MODVOLC		RST
	AVHRR no. (%)	MODIS no. (%)	AVHRR no. (%)	AVHRR no. (%)	MODIS no. (%)	MODIS no. (%)
Images	289	63				
Correct			130 (45%)	78 (27)	38 (60)	52 (82)
Missed			159 (55)	211 (73)	25 (40)	11 (16)
False			17 (12)	1 (1)	0 (0)	1 (2)
Pixels	2,377	384				
Correct			438 (18)	291 (12)	157 (41)	310 (81)
Missed			1,939 (82)	2,086 (88)	227 (59)	74 (22)
False			82 (15)	2 (0.7)	0 (0)	31 (9)

relatively low temperature of 50°C to 60°C (Harris et al. 1995), band 3 saturates over nearly all lava pixels, thereby maxing out. This allows the TIR (band 4) pixel-integrated radiances to “catch up” with the maxed-out MIR values, thereby reducing the NTI to below the threshold value. At times, band 3 also responds to the hot spot by recording anomalously low temperatures (of around −30°C) just down scan of the anomaly, thereby exacerbating the problem (Fig. 13c–d). Therefore, the most anomalous pixels in the AVHRR images cannot be detected by MODVOLC because the MIR radiances are lower than the band 4 radiances resulting in an NTI which is much lower than the assigned threshold of −0.8. This again is not a problem with the algorithm design, but is due to limitations of the AVHRR data.

Intermittent strombolian activity was also difficult to detect using MODVOLC. When MODVOLC used MODIS data only 13% of the images and 16% of the total pixels were correctly identified as anomalous. The reason for this incidence of detection during normal activity at Stromboli was due to the high threshold set (Fig. 13g–h). The radiance required to create a NTI high enough to exceed the threshold (Fig. 6) requires a suitably large and hot

source, which Stromboli does not always offer. For example, one 4-m-wide vent at 950°C gives a band 22 radiance of $0.33 \text{ Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$, and a band 32 radiance of $8.5 \text{ Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$, hence, an NTI of −0.92. Again, more pixels would be flagged if a lower threshold was set, but the likelihood of false detections would increase. MODVOLC run using AVHRR data at Stromboli during 2006 also performed poorly. This time the low detection rate was not caused by the MIR data saturating (band 3 rarely saturates over a target of such a radiance), but instead due to the high threshold. Most of the NTI's produced from anomalous pixels ranged between −0.85 and −0.88, a value not great enough to trigger an alert. In order to alleviate this problem a slightly lower NTI could be set. However, the benefit of a poor detection incidence is a zero rate of false detection.

MODVOLC detected 56% of the images that contained hot spots during the Augustine eruption. Again, the low rate of positive identification resulted from the occurrence of lower intensity anomalies associated with lava dome extrusion (Fig. 6). Likewise, the percent of thermally anomalous pixels detected by MODVOLC was low at 34%, so that only a fraction of all anomalous pixels within

Table 8 Number of correct, missed, and false detections during background activity at Vulcano 2006

Algorithm	Manual		VAST	MODVOLC		RST
	AVHRR no. (%)	MODIS no. (%)	AVHRR no. (%)	AVHRR no. (%)	MODIS no. (%)	MODIS no. (%)
Images	0	0				
Correct			n/a	n/a	n/a	n/a
Missed			n/a	n/a	n/a	n/a
False			58 (18)	n/a	n/a	n/a
Pixels	0	0				
Correct			0	n/a	n/a	n/a
Missed			0	n/a	n/a	n/a
False			106 (100)	n/a	n/a	n/a

Table 9 Number of correct, missed, and false detections during at Lipari 2006

Algorithm	Manual		VAST	MODVOLC		RST
	AVHRR no. (%)	MODIS no. (%)	AVHRR no. (%)	AVHRR no. (%)	MODIS no. (%)	MODIS no. (%)
Images	0	0				
Correct			n/a	n/a	n/a	n/a
Missed			n/a	n/a	n/a	n/a
False			27 (8)	n/a	n/a	n/a
Pixels	0	0				
Correct			0	n/a	n/a	n/a
Missed			0	n/a	n/a	n/a
False			39 (100)	n/a	n/a	n/a

each hot spot images were detected (Fig. 13i–j). Again this is due to the NTI threshold used by MODVOLC. In order to correctly identify hot spots at lava domes, the threshold would have to be lowered.

During 2006, MODVOLC did not produce any alerts for Vulcano and Lipari. This confirms the robustness of

MODVOLC (when used as a global threshold) for returning minimal false detections. The question thus begins to boil down to what is more important: genuine detection with no false detections but many missed low intensity events, or a 100% detection incidence with many false events?

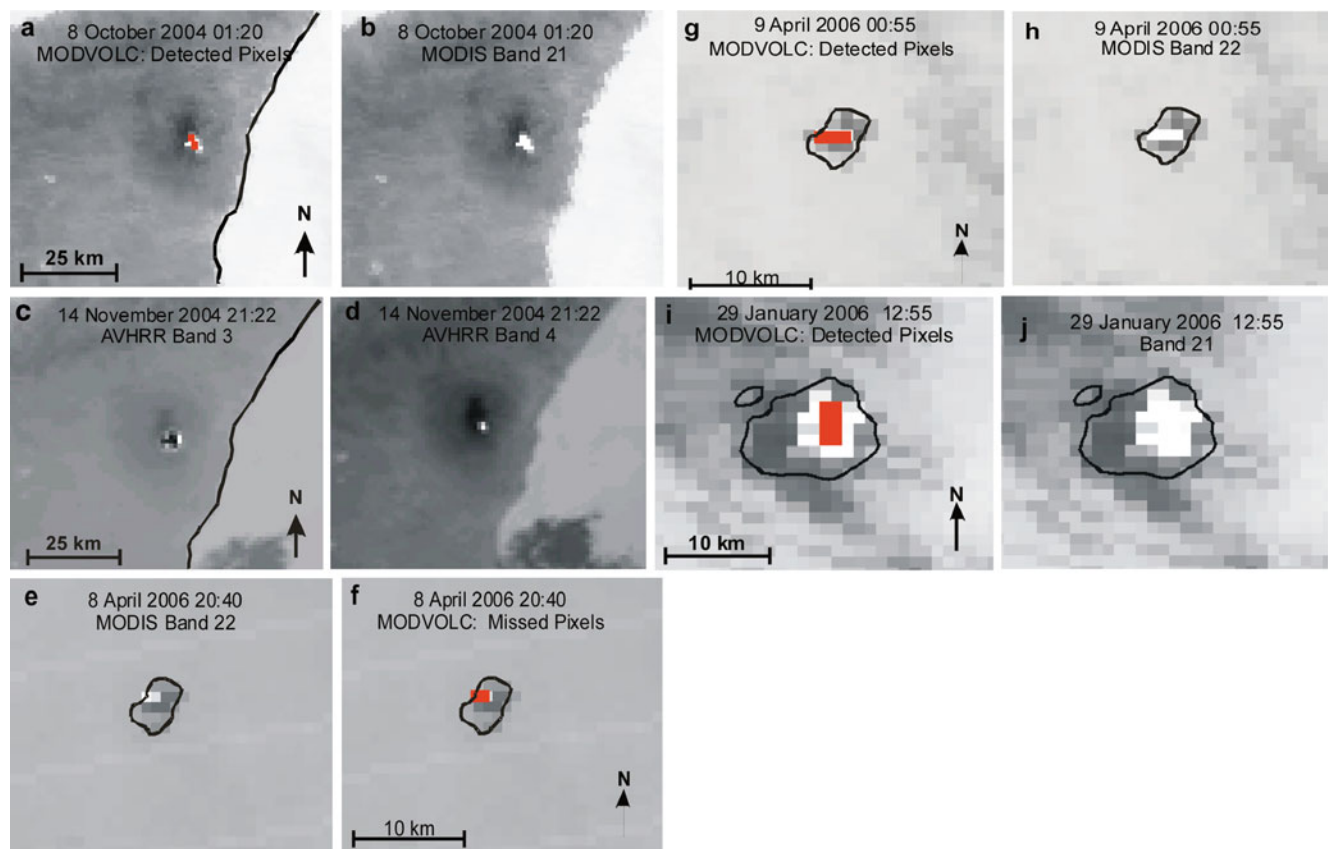


Fig. 13 Examples of volcano hot spot images with pixels flagged as anomalous by MODVOLC. (a, b) Good result at Etna: eight out of the twelve anomalous pixels are flagged. (c, d) AVHRR image acquired during the Etna 2004–2005 eruption with no hot spot pixels detected. Note that there are six pixels with anomalously low temperatures (black pixels). These are ‘recalibration’ pixels (see text for explanation).

(e–h) Example of thermal anomaly missed (e, f) and found at Stromboli (g, h). In the missed case, the anomaly is apparent in the MIR, but not large or hot enough to raise the pixelintegrated temperature in the TIR. (i, j) Example of thermal anomaly at Augustine where only a portion of the pixels were detected by MODVOLC

Robust Satellite Technique

RST performed well when applied to effusive activity, detecting 71% of the anomalous images and 45% of the pixels within those images during Etna's 2006 eruption (Fig. 14a–b). RST was not as effective at detecting the lava flows during the Stromboli 2007 effusive eruption. The algorithm only flagged 38% of the hot spot images and detected 26% of the anomalous pixels within those images. These percentages are unexpected given the high radiances

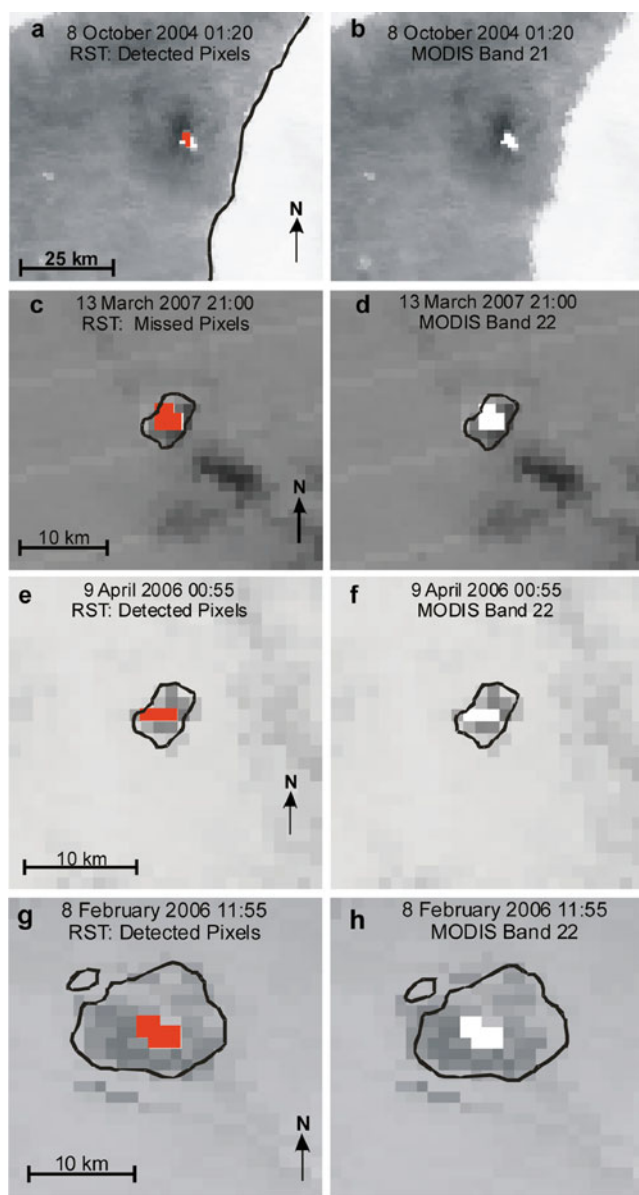


Fig. 14 Examples of volcano hot spot images with pixels flagged as anomalous by RST. (a, b) Good result at Etna: six out of the twelve anomalous pixels are flagged. (c, d) Example image acquired during the Stromboli 2007 eruption where RST identified none of the anomalous pixels. (e, f) Example image acquired at Stromboli where RST identified all of the anomalous pixels. (g, h) Example image acquired during the Augustine 2006 eruption where RST identified all of the anomalous pixels

produced during effusive eruptions (Fig. 14b). However, the algorithm did not produce any false detections during the two eruptions.

When applied to strombolian activity, RST flagged 44% of the hot spot images and detected 49% of the anomalous pixels within the images (Fig. 14c–d) with a false detection incidence of 21%. RST's performance during the 2006 eruption of Augustine was the best of any of the algorithms. RST flagged 82% of the hot spot images and detected 81% of the anomalous pixels within those images. Therefore RST detected most of the anomalies that occurred during the eruption period and detected most of the pixels within those images (Fig. 14g–h). The occurrence of false detection was also low, with RST producing just one false detection during that entire 3-month period. The excellent performance of RST during the Augustine eruption might be due to the location and time of year that the eruption occurred: Augustine having a high latitude and the eruption occurring in winter. Background temperatures in Alaska will thus be much lower than those in the Mediterranean. Normal background temperatures in December frequently are below 0°C, therefore the anomalous activity occurring at the dome would be more prominent than it would be if the background temperatures were warmer. Therefore using the means of the very-low-temperature background pixels it is easier to detect lower temperature thermal anomalies.

RST experienced no false detections at Vulcano during 2006 and only one false positive at Lipari. This false positive occurred near the coast and could have been caused by a low level anthropogenic (fire) source.

Discussion II: algorithm applicability

Each algorithm has different merits depending on user requirements. For example, for a globally based search, algorithm processing time and data volume may be an issue. Sensitivity, however, may not be so important, i.e., it is accepted that subtle hot spots will be missed to limit the amount of false detections and allow capture of the main events for construction of a global inventory and overview of volcanic activity. A local or regional application, on the other hand, may be able to ingest greater amounts of data and operate with higher precision. The smaller area considered limits the amount of data needed, and monitoring of a single volcano or region may allow a large number of images to be stored and used, and still processed in a timely fashion. In addition, more subtle, regionally applicable, thresholds can be set (e.g., MODLEN). This may result in high false detection rates. However, these can be manually checked and rejected (as can be done in VAST), where in a monitoring setting it is better to have some false alerts (whose validity can be assessed by the operator) than to miss an eruption.

Determining whether or not an algorithm can be run on large areas versus its ability to detect small and subtle anomalies is a key issue. A potential user thus needs to weigh the strengths and weaknesses of the various algorithms and determine which of the algorithm best fits their needs. To assess the applicability of each algorithm type to achieve a global or local monitoring objective, we ranked the algorithms from zero to one based on (1) the amount of data required to run the algorithm, (2) the processing time required to run the algorithm, (3) the global applicability of the method, (4) the local precision of the algorithm, and (5) the subtlety of anomalies the algorithms could detect (Table 10). While the first two specifications will allow a potential user to determine if they have enough storage and computing capacity to efficiently run the algorithm, the latter three tests allow the user to determine if the algorithm meets the group's needs in terms of global, regional, or local application.

When considering the amount of data required to run the algorithm, VAST and MODVOLC are both assigned one and RST a zero. This is because both MODVOLC and VAST can run on an image by image basis, thereby being executable even if only a single image is available. Therefore large amounts of data or access to a data archive are not needed to obtain accurate results. In order for RST to obtain the most accurate results, it must have an archive of data to obtain the necessary temporal means and standard deviations on which the algorithm is based. A year's worth of MODIS data for one location, is ~150 GB, with file sizes growing once the images are georectified (a MODIS image increasing in size from 65 to 350 MB once georeferenced, so that the total size for the 1-year data set increases to ~750 GB). Therefore the data required to run the algorithm requires a matching storage space.

Processing time is dependent upon the amount of data that it takes to run the algorithm, therefore VAST and MODVOLC again both received an assignment of one. It takes minimal time to run the algorithms for one image (<1 min, typically a few seconds), but the requirement of RST to examine a large amount of data to run the algorithm requires longer run and set-up times. It takes between 12

and 72 h to download a year worth of nighttime MODIS data for one location, depending upon the download speed. In addition, it takes around three hours to georectify the data, plus around 2 h to actually run the algorithm. Therefore the total set-up time for RST is between 18 and 72 h. This, of course, is dependent upon the processor speed. It should be noted that although the initial set-up time for RST is much higher, once the archive of data is obtained, and the data is processed, the upkeep time for the algorithm decreases. For example, once the means and standard deviations are determined for a given location, the algorithm can run just as fast as the other two algorithms. Therefore for initial set-up RST would be assigned a zero, but for running continuously RST would receive a one.

Next, global applicability is assessed. Because MODVOLC was designed as a global algorithm, it runs well in a global environment as already shown by Wright et al. (2002; 2004), and therefore, it was assigned a one. RST and VAST, on the other hand, are not as easily applied globally and were both assigned zero. The areas that would need to be defined and applied for VAST to operate globally would be numerous. There are over 600 historically active volcanoes in the world (Simkin and Siebert 1994; Siebert et al. 2010) and even if all of those were targeted, activity at previously inactive volcanoes could not be detected. RST on the other hand would require a large amount of data to run on a global dataset and is currently impractical. A total of 576 images are collected across the globe on any given day which results in a total data set size of 62 GB. Therefore to acquire an archive of just 1 year of data requires 22 TB of disk space. For more accurate results, over a year's worth of data would have to be amassed with the total disk space required to house that data increasing by 22 TB per year.

Local precision and subtlety of the hot spot that can be detected by the algorithms have an inverse relationship with the global applicability. If an algorithm runs effectively on a global scale, it will have a high threshold which limits the number of false detections but precludes detection of subtle anomalies. For the same reason, such algorithms find it difficult to detect all of the pixels that make up a hot spot, such as the low intensity pixels around the edge of an anomaly, or more subtle hot spots associated with strombolian or lava dome activity. Therefore MODVOLC was assigned a zero for both the local precision and subtlety of anomalies that can be detected. RST and VAST on the other hand work off the information for the area to which they are applied and were designed to detect hot spots at specific locations. Therefore they have lower, more sensitive, thresholds designed specifically for a given location or, in the case of VAST, taken from the image and region being considered. Therefore both VAST and RST were assigned one for local precision. When considering the subtlety of

Table 10 Ranking of the algorithm applicability to operational monitoring

Criteria	VAST	MODVOLC	RST
Amount of data	1	1	0
Processing time	1	1	1
Global applicability	0	1	0
Local precision	1	0	1
Subtlety of anomaly	0	0	1
Total	3	3	3

anomaly that can be detected RST was assigned one and VAST was assigned zero. Because RST operates using the MIR radiances, it can therefore detect anomalies caused by the smaller, less intense anomalies compared to VAST which currently operates using the TIR which only detects the larger and more intense anomalies that cause increases in the TIR (Fig. 6). Combining all of these criteria we obtain a dead heat, i.e., all algorithms receive an overall score of three (Table 10). To calculate the algorithm's overall performance, the average percentage of false positives were subtracted from the average percentage of correctly identified anomalous images so that, if the algorithm performed perfectly, it would have a score of 100. This number was then weighted by the qualitative score in Table 10 (but because each of the algorithms received the same score, this step is not really necessary). However, a prospective user may want to apply a different weight depending on needs. For example, if the requirement is for a local check but with the availability of just one image, Table 10 matrix can be used to assign a weight of two for VAST, one for MODVOLC and zero for RST.

Using our quantitative tests, VAST had an average detection incidence of 73% and a false detection incidence of 21%, so that the overall accuracy rating of the algorithm for the four test cases was 54 (Table 11). MODVOLC was assessed based on its performance using AVHRR and MODIS imagery separately. When using AVHRR it has an average detection incidence of 31% and a false detection incidence of 3%, leading to an overall accuracy rating of 28 (Table 11). When using MODIS, MODVOLC had an average detection incidence of 64% and a false detection incidence of 3%, resulting in an accuracy rating of 61 (Table 11). RST had a detection incidence of 59% with a false detection incidence of 6% to result in a total accuracy rating of 53 (Table 11). Again, the prospective user can use the statistics given in Tables 4, 5, 6, 7, and 8 to make an assessment as to the suitability of each algorithm to the case in hand depending on the detection problem: lava flows, open vents, lava domes, or fumaroles.

Adaptations to algorithms

Adaptations to all of the algorithms could be made to make the algorithms perform well under all circumstances; adaptations can also be made depending upon the data

type used to improved performance. To do this, our recommendations are as follows (see also Tables 12 and 13).

Adaptations explicitly for VAST include reverting to the original ΔT algorithm for cases where saturation of the MIR band does not occur. This would allow improved detection of low-temperature anomalies, such as those observed at Stromboli, Augustine, and Vulcano. Reintroducing the ΔT technique would also filter out the occurrence of false detections that occurred due to strong temperature contrasts at boundaries between two different land types (i.e., snow vs. lava fields and island vs. sea). If saturation does occur in the MIR band, then the TIR band can be utilized—as in this study.

Potential adaptations to MODVOLC include threshold modification based upon the location of interest. MODVOLC was designed for global applicability and therefore has a high threshold to reduce the occurrence of false detections. If a user needs to apply the algorithm to a specific area of interest it would be easy to implement a new threshold to detect thermal anomalies for that location, where the steps described in Wright et al. (2004) can be followed to determine an appropriate NTI for a specific location, thereby allowing hot spots in that region to be more fully detected, while still limiting the number of false detections.

RST could be improved by insuring that the mean and standard deviations are for periods when no activity is occurring. An increase in activity at the volcano will result in the mean being higher than ambient and therefore will not allow more subtle anomalies to be detected. RST can also be evaluated to see what standard deviation should be used for each location; this would allow more subtle anomalies to be detected. For example a lower standard deviation might be needed at a location where the normal background temperatures are higher (e.g., Stromboli), but do not have to be set as high for areas where the background temperatures are lower (e.g., Augustine).

Detection and higher level applications

When evaluating an algorithm's performance it is important to first distinguish how the output will be used. If an algorithm is used to simply alert a user to the presence of a hot spot, then whether or not the algorithm identifies all of

Table 11 Ranking of algorithms performance based on quantitative and qualitative assessments

Algorithm	% Correct	% False	% Correct–% false
VAST	73	21	52
MODVOLC (AVHRR)	31	3	28
MODVOLC (MODIS)	64	3	61
RST	59	6	53

Table 12 Suggested adaptations to algorithms when using AVHRR data to increase accuracy of the algorithms for the various types of volcanic activity

AVHRR DATA											
VAST				MODVOLC				RST			
Flows	Strom.	Domes	Fumaroles	Flows	Strom.	Domes	Fumaroles	Flows	Strom.	Domes	Fumaroles
Georeference Data				Georeference Data				Georeference Data Errors > 2 km?			
Cloud Detection T4> 10°C				Cloud Detection T4> 10°C				YES		NO	
No Quantitative Analysis				No Quantitative Analysis				Abandon		Continue	
Use T4	Use ΔT	Use ΔT	Use ΔT	T3 Saturated				Cloud Detection T4> 10°C			
				YES				No Quantitative Analysis			
				Abandon				Use T4	Use T3	T3	UseT3
				NO				Saturated?			
				Continue						NO	YES
										Use T3	Use T4

the anomalous pixels is inconsequential. It is sufficient that an alert is issued notifying the user to the presence of the hot spot in as timely a manner as possible. In this regard, direct data reception, immediate processing and application of an algorithm that will find the hot spot are required to minimize transmission delays (Harris et al. 2002a). Thus, the algorithm with the highest return of detected hot spots is preferential.

However, if information from the alert is to be used for quantitative analysis, the number of detected pixels may be a

problem if some are missed. Automated detection routines have been used to obtain accurate radiance time series (e.g., Wright et al. 2005; Patrick et al. 2005; Rothery et al. 2005; Harris and Ripepe 2007a; Lyons et al. 2010; van Manen 2009). The output have also been used to convert to total anomaly heat flux or lava discharge rate (e.g., Wright and Flynn 2004; Ripepe et al. 2005; Hirn et al. 2008; Coppola et al. 2009; Vicari et al. 2009; Delle Donne 2011). Such applications require that all pixels (or at least the most radiant pixels containing the majority of the active lava flow field radiance)

Table 13 Suggested adaptations to algorithms when using MODIS data to increase accuracy of the algorithms for the various types of volcanic activity

MODIS DATA											
VAST				MODVOLC				RST			
Flows	Strom.	Domes	Fumaroles	Flows	Strom.	Domes	Fumaroles	Flows	Strom.	Domes	Fumaroles
Georeference Data				Georeference Data				Georeference Data Errors > 2 km?			
Cloud Detection T32>-10°C				Cloud Detection T32>-10°C				YES		NO	
No Quantitative Analysis				No Quantitative Analysis				Abandon		Continue	
Use ΔT	Use ΔT	Use ΔT	Use ΔT					Cloud Detection T32>-10°C No Quantitative Analysis			
								T22 Saturated?			
								NO		YES	
								Use T22		Use T21	

are detected. Thus, the algorithm that gives highest return of detected anomalous pixels is preferential for such cases.

We, here, examine the effects of not detecting all of the anomalous pixels when calculating radiative power loss for two different types of thermal anomaly. Two test cases are examined: first a high intensity thermal anomaly associated with lava flow activity during Etna's 2004–2005 eruption. The second is a low intensity anomaly associated with lava dome growth during Augustine's 2006 eruption. Radiative power loss, for a given wavelength, can be calculated using (Kaufman et al. 1998):

$$E_f = 4.34 \times 10^{-19} (T_h^8 - T_b^8) \quad (5)$$

where T_h (in K) is the pixel-integrated temperature of the anomalous pixel and T_b is the temperature of the surrounding ground. Total radiative power loss is then calculated by summing the power from all of the anomalous pixels.

Large, high-temperature anomalies (such as those caused by extensive lava flows) are usually composed of a core of highly radiant pixels surrounded by a zone of lower radiance pixels which contribute a relatively small percentage to the total radiance. For example, the MODIS image acquired during for Etna on 8th October 2004 at 1:20 UTC (Fig. 12a) was composed of ten anomalous pixels which gave a total radiative power loss of 905 Wm^{-2} . MODVOLC detected six (60%) of these pixels, which accounted for 797 Wm^{-2} or 88% of the total radiative power loss. RST detected four (40%) of the anomalous pixels whose radiances accounted for 616 Wm^{-2} or 68% of the total power loss. Therefore even though MODVOLC and RST missed 40% and 60% of the pixels, respectively, this only resulted in an underestimation of the total power loss by 12% and 32%, respectively.

Lower temperature anomalies, such as those associated with lava dome growth, often have a smaller percentage of highly radiant pixels and a larger percentage of lower radiance pixels. An example of such a case is the MODIS image acquired for Augustine on 8th February 2006 at 11:55 UTC. The thermal anomaly was composed of 10 pixels with a total radiative power loss of 176 Wm^{-2} . MODVOLC detected four (40%) of these pixels whose radiances accounted for 80 W or 50% of the total power loss. RST, on the other hand, detected all 10 pixels and therefore accounted for 100% of the total power loss. In this case, nondetection of 60% of the pixels by MODVOLC resulted in an underestimation of the total radiative power loss by 50%, a result of the radiance being distributed more evenly across the entire anomaly in the low intensity case.

Therefore, the size and temperature of the feature will have an effect on whether or not the total number of pixels detected by an algorithm, and the incidence of missed pixels, will greatly influence quantitative results. For more intense anomalies, even if a substantial percentage of lower

intensity pixels are not detected by the algorithm, non-inclusion of the low radiance pixels will not have a great effect if those pixels are much less radiant than those detected. On the other hand, if an algorithm misses the same percentage of pixels across a lower intensity anomaly, the effect is greater because a greater proportion of total radiant power loss will be missed.

Conclusions

The repeat coverage of all earth surface points guaranteed by a satellite-based sensor makes space-based hot spot detection of value in monitoring volcanoes on a global scale. However, there is often just too much data to check manually. GOES, for example, provides 96 images per day for all volcanoes within the latitudes of North America, analysis of which requires some degree of automation. As a result, the use of automated algorithm output is the only plausible means to detect all hot spots across geographically extensive regions, construct global or regional heat flux inventories, or to provide timely radiance time series during eruptions. Defining the detection algorithms available to the remote sensing volcanologist, their capabilities and performance is therefore essential if we are to use, apply and trust such data sets.

In applying each of the main algorithms used in volcanology (VAST, MODVOLC, and RST), we find that there is a tradeoff between catching all of the hot spots present in a series of images and the rate of false detection. While an algorithm that has a low occurrence of false detections will be more likely miss subtle anomalies, those that detect subtle anomalies will have a higher rate of false detections. As the number of correctly identified hot pixels increases, so too does the occurrence of false detections. This effect was most pronounced for VAST, where for Stromboli 2006 and Augustine 2006 the number correct detections was the greatest out of all of the algorithms, but the number of false detections was also the greatest.

Of course, each algorithm performs well under the circumstances that they were designed for. VAST was developed to detect intense anomalies caused by active lava flows, and suits cases where MIR data cannot be used to detect thermal anomalies due to sensor saturation. MODVOLC was designed to detect thermal anomalies on a global scale with a minimal occurrence of false positives, having a high threshold that detects the most intense anomalies, but therefore misses some of the less intense anomalies. With the exception of the 2007 Stromboli effusive eruption, RST probably performed the best of all algorithms tested here, but requires exacting geolocation requirements, the availability of an extensive data archive for the volcano in question and can place heavy demands

computer resources. Thus while MODVOLC is an excellent option for efficient analysis of global data sets, VAST and RST are more useful for regional or local applications. We note, though, that MODVOLC is best adapted and/or combined with VAST or RST to allow improved regional application to smaller, site specific cases as done by, for example, MODLEN (Kervyn et al. 2006), MYVOLC (Hirn et al. 2008), and MODVOLC2 (Koeppen et al. 2011).

It is vital that the users determine their needs before selecting an algorithm. For a volcano observatory, it may be more important to detect all hot spots present, with the inconvenience of being sent false alerts outweighing the cost of missing a potentially important, but subtle, hot spot. On the other hand, if one wants to track global or regional volcanic activity trends, the more subtle anomalies may not be as important in contributing to the total heat flux inventory and can thus be discarded. Missing a few subtle hot spots outweighs the problems induced by ingestion of a large number of false alerts that need to be filtered from the time series. An algorithm should thus be used to flag cases for further user inspection, and manual checking should always be completed. In cases where data are to be used for precise hot spot mapping and further quantitative applications, flagged images must be checked to ensure that all anomalous pixels have been located, falsely located pixels will also need to be removed if precise quantitative assessments are to be made.

No algorithm performs with 100% accuracy and, within current data constraints, none ever will be: all can miss hot spot cases and can generate false detections. However users are now in a position to run a range of algorithms capable of aiding the operator in fast and efficient identification of volcano hot spots in satellite thermal data. These algorithms are especially useful in the analysis of large (in a temporal and spatial sense) data sets. Our intent is that the algorithm performance statistics given for a range of case type eruption scenarios will aid in selection of an appropriate algorithm for monitoring and/or research applications. As the storage capacities and the processing times of computers increase the amount of data that can be used increases and processing times decrease. Therefore, with time, running multiple algorithms is becoming easier, so that combinations or all of the end-member algorithms could be run in parallel. Satellite remote sensing and the use of automated algorithms is a growing operational field in volcanology which, when coupled with other data streams (seismic, geodetics, historical data) will make monitoring faster, more accurate, and comprehensive.

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